Chapter 1: Numerical Methods

In this chapter, we introduce a locally mass conserving numerical framework. A conforming mixed finite element method (MFEM) is proposed to solve the static Darcy-Stokes coupled methanics system. The proposed conforming MFEM captures the normal fluxes across element edges, ensuring compatibility with local mass conservation. The transport system is solved using a finite volume method (FVM), which inherently ensures local mass conservation.

We begin by introducing the mesh partition used for both the MFEM and the FVM schemes. Based on the mesh partition, two stable element pairs are presented, one for Darcy flow and one for Stokes flow. We then present a noval nonlinear WENO weighting scheme for the FVM. Error estimates are provided for both the MFEM and the FVM schemes.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$, where $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. For a given $h > 0$, let $\mathcal{T}_h$ be a *triangulation* of the closure of the domain $\overline{\Omega}$ made of quadrilaterals E, i.e.,

$$\overline{\Omega} = \cup_{E \in \mathcal{T}_h} E, \quad h_E < h, \tag{1.1}$$

where two quadrilaterals are either disjoint or share one edge or one vertex. We assume that the quadrilateral mesh is uniformly shape regular. A formal definition of uniform shape regularity of a quadrilateral mesh is given in Girault and Raviart (2011).

Definition 1.1. For any quadrilateral $E \in \mathcal{T}_h$, form the triangle $T_i, \quad i \in \{0, 1, 2, 3\}$ by excluding vertex v_i from the quadrilateral element E, and use the remaining three vertices. Let

$$
\begin{aligned}
h_E &= \text{diameter of Element } E, \\
\rho_E &= 2 \min_{1 \le i \le 4} \{\text{diameter of largest circle inscribed in } T_i\}.
\end{aligned} \tag{1.2}
$$

A mesh partition $\mathcal{T}_h$ is considered to be uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that for all quadrilaterals $E \in \mathcal{T}_h$,

$$
\rho_E / h_E \ge \sigma_* > 0. \tag{1.3}
$$

Note that no subtriangle T_i of E has zero measure, i.e., each quadrilateral E is assumed to be strictly convex, not degenerating into a triangle, line or point.

The quadrilateral $E \in \mathcal{T}_h$ is referred to differently depending on the context. In the finite element method, a quadrilateral E is called an *element*, while in the finite volume method, it is typically referred to as a *cell*. For simplicity, we use the term *element* throughout this section.

In Figure 1.1, we show a reference element $\hat{E} = [-1, 1]^2$, a local reference element $\tilde{E}$ and a general quadrilateral element E. The edges of element E are denoted by e_i, $i \in \{0, 1, 2, 3\}$, and the vertices of the element E are denoted as v_i, $i \in \{0, 1, 2, 3\}$, both numbered in the counterclockwise order. We begin counting from zero, first to align with the indexing convention of C++, and second to facilitate representation in terms of modulo arithmetic, which also starts from zero. The vertices can be understood by $v_i = e_i \cap e_{i+1}$, where $i \equiv i \pmod 4$. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outwards from the element E. The corresponding unit tangent $\boldsymbol{\tau}_i$ is obtained by rotating $\boldsymbol{\nu}_i$ counterclockwise, following the right-hand rule.

We assume the general quadrilateral element E can be obtained by scaling from the local reference element $\tilde{E}$ such that $\tilde{E} = E/H$, where H is the maximal edge length of E, i.e., the length of edge e_1 in Figure 1.1. Here, $|e_1| = H$ and $|\tilde{e}_1| = 1$. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to \tilde{E}$ mapping the vertices of $\hat{E}$ to the vertices of $\tilde{E}$. Note that the unit normal and the unit tangent vectors are invariant under the scaling mapping H.

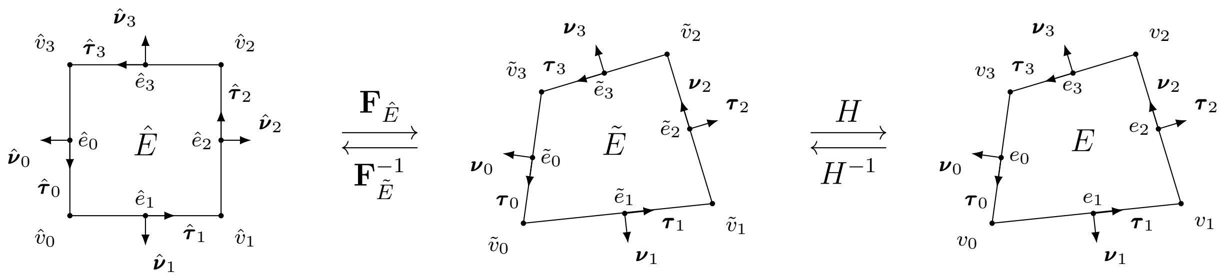


Figure 1.1: An exhibition of a reference square element, a local reference element, and a general non-degenerate quadrilateral element.

For the ease of implementation, we use a logically rectangular mesh, where each element is indexed using Cartesian coordinates $\{i, j\}$. In Figure 1.2, we illustrate a full logically rectangular, quasi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distorting the interior vertices of a square mesh. The boundary vertices $v_i \in \partial\Omega$ remain fixed and are not subject to distortion.

1.2 Mathematical Preliminaries

In this section, we introduce some mathematical concepts that are relied upon in the remainder sections. The theory and notations are presented in a general dimension sense, but in practice, our discussion is confined to two dimension cases. In addition, for simplicity, we restrict our consideration to real-valued functions.

To begin with, we recall the definition of a Sobolev space.

Definition 1.2. Let $\Omega \subset \mathbb{R}^d$ be a domain, $1 \le p \le \infty$, and $m \ge 0$ be an integer. The Sobolev space of m derivatives in $L^p(\Omega)$ is

$$W_p^m(\Omega) = \{f \in L^p(\Omega) : D^\alpha f \in L^p(\Omega) \text{ for all multi-indices } \alpha \text{ such that } |\alpha| \le m\}, \tag{1.4}$$

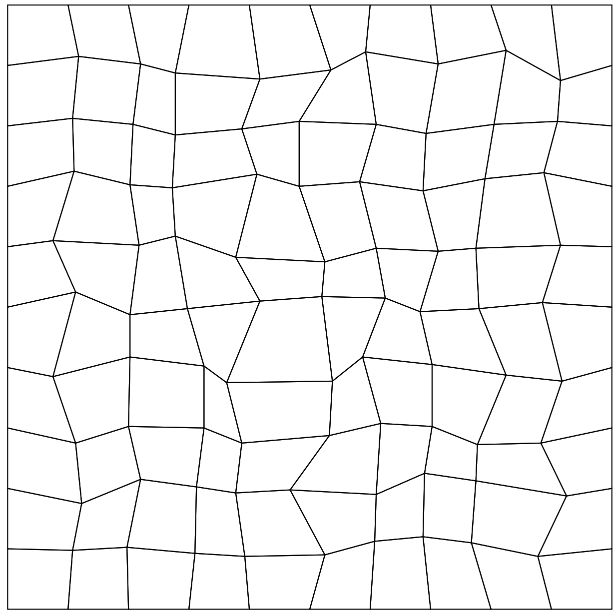


Figure 1.2: An exhibition of a full logically rectangular quasi-uniform quadrilateral mesh $\mathcal{T}_h$.

where the multi-index $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_{N-1}) \in \mathbb{N}^N$ and

$$|\alpha| = \sum_{i=0}^{N-1} \alpha_i.$$

The multi-index derivative is defined as

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_0^{\alpha_0} \ldots \partial x_{N-1}^{\alpha_{N-1}}}.$$

For $f \in L^p(\Omega)$, the $L^p(\Omega)$ norm is

$$\|f\|_{L^p(\Omega)} = \begin{cases} \left(\int_\Omega |f|^p \right)^{1/p}, & p < \infty, \\ \operatorname*{ess\,sup}_{x \in \Omega} |f(x)|, & p = \infty. \end{cases} \tag{1.5}$$

We proceed with the definition of the Sobolev norm and seminorm.

Definition 1.3. For $f \in W_p^m(\Omega)$, the norm $\|\cdot\|_{W_p^m}$ is

$$\|f\|_{W_p^m(\Omega)} = \left\{ \sum_{|\alpha| \leq m} \|D^\alpha f\|_{L^p(\Omega)}^p \right\}^{1/p}, \quad p < \infty, \tag{1.6}$$

and

$$\|f\|_{W_\infty^m(\Omega)} = \max_{|\alpha|\le m} \|D^\alpha f\|_{L^\infty(\Omega)}, \quad p = \infty. \tag{1.7}$$

The seminorm $|\cdot|_{W_p^m}$ is

$$|f|_{W_p^m(\Omega)} = \left\{ \sum_{|\alpha|=m} \|D^\alpha f\|_{L^p(\Omega)}^p \right\}^{1/p}, \quad p < \infty. \tag{1.8}$$

In the special case when $p = 2$, we denote $W_2^m(\Omega) = H^m(\Omega)$. The corresponding norm and seminorn are then written as $\|\cdot\|_{H^m(\Omega)}$ and $|\cdot|_{H^m(\Omega)}$ respectively. Recall the Trace Theorem in $H^m(\mathbb{R}^d)$.

Theorem 1.1. *(Trace Theorem)*

Let k and d be integers with $0 < k < d$. The restriction map R extends to a bounded linear map from $H^m(\mathbb{R}^d)$ onto $H^{s-k/2}(\mathbb{R}^{d-k})$, provided that $s > k/2$.

Proof. See Arbogast and Bona (2025). □

Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ defined on $\mathbb{R}^d$ can be calculated as

$$\dim \mathbb{P}_r(\mathbb{R}^d) = \binom{r+d}{d} = \frac{(r+d)!}{r!d!}. \tag{1.9}$$

For simplicity, we momentarily restrict $d = 2$. Let $\mathbb{Q}_{r,s}(\omega) :- \mathbb{P}_r(\omega) \otimes \mathbb{P}_s(\omega)$ denote the space of tensor product polynomials derived by polynomials of degree r in x and s in y on $\omega \subset \mathbb{R}^2$. The dimension of $\mathbb{Q}_{r,s}$ is

$$\dim \mathbb{Q}_{r,s}(\mathbb{R}^2) = \dim \mathbb{P}_r(\mathbb{R}) \times \dim \mathbb{P}_s(\mathbb{R}) = (r+1) \times (s+1). \tag{1.10}$$

We refer to the Bramble-Hilbert lemma for basic interpolation error estimation for polynomials of degree r.

Lemma 1.2. *(Bramble-Hilbert Lemma)*

Let $\Omega \in \mathbb{R}^d$ be a domain, and function $f \in W_p^{k+1}(\Omega)$, $k \geq 0$, $1 \leq p < \infty$. There exists a constant C such that

$$\inf_{p \in \mathbb{P}_k(E)} \|f - p\|_{W_p^m(E)} \leq C h^{k+1-m} |f|_{W_p^{k+1}(E)}, \quad m = 0, 1, \tag{1.11}$$

where the constant C has an upper bound over all elements $E \in \mathcal{T}_h$ due to the shape regularity assumption.

Proof. See Bramble and Hilbert (1970) and Dupont and Scott (1980b). □

1.3 Direct Serendipity Space

Serendipity finite elements have the same accuracy of approximation with classical tensor product elements while using fewer degrees of freedom. However, the classical serendipity spaces suffer from poor approximation on quadrilaterals. To address this issue, we follow the approach of Arbogast et al. (2022) by defining local shape functions directly on quadrilateral elements, thereby recovering the desired accuracy. In the next sections, H^1 and $H(\mathrm{div})$ conforming direct mixed finite element spaces are constructed with the direct serendipity function spaces.

To begin with, we introduce some auxiliary distance functions. We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{1.12}$$

where $\mathbf{x}_i^* \in e_i$ is any point on edge e_i. We pick $\mathbf{x}_i^* = \mathbf{x}_{v,i}$ for simplicity, where $\mathbf{x}_{v,i}$ denotes the coordinates of corner v_i. The distance function λ_i returns a positive value as long as the point is located in the interior of the element E.

We define two additional distance functions giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to the diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\boldsymbol{\nu}_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by rotating counterclockwise of the unit vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}^*_{d,i}) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 0, 1, \tag{1.13}$$

where $\mathbf{x}^*_{d,i}$ is any point on the diagonal line d_i. Here, we use the corner point v_i for simplicity, i.e., $\mathbf{x}^*_{d,i} = \mathbf{x}_{v,i}$.

We now present a set of nodal basis functions for the first and second order direct serendipity spaces, $\mathcal{DS}_1$ and $\mathcal{DS}_2$.

1.3.1 Second-order Space $\mathcal{DS}_2$

The dimension of the second-order polynomials in two dimension is calculated by (1.9) as

$$\dim\mathbb{P}_2(\mathbb{R}^2) = \binom{2+2}{2} = \frac{(4)!}{2!2!} = 6, \tag{1.14}$$

while a quadrilateral element requires a minimum of 4 vertex dofs and 4 edge dofs. Therefore, we supplement polynomial space $\mathbb{P}_2(E)$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$, spanning the supplemental function space $\mathbb{S}_r^{\mathcal{DS}}(E) = \operatorname{span}\{\phi_{s,1}, \phi_{s,2}\},\ \ r \geq 2$. We present the second-order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}_2^{\mathcal{DS}}(E). \tag{1.15}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{0, 1\}. \tag{1.16}$$

We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.17}$$

which satisfies the conditions

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{1.18}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad i \in \{0,1,2,3\}, \tag{1.19}$$

where $i \equiv i \pmod 4$ as usual, and $R_{i,i+2} = -R_{i+2,i}$ according to the equation (1.17). Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.3, we draw R_0 on a general quadrilateral element, which evaluates to 1 on edge e_0, and evaluates to 0 on the opposite edge e_2.

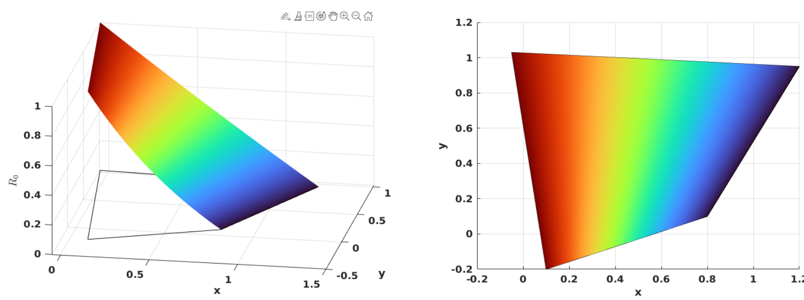


Figure 1.3: Rational function R_0.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^{(2)}_{e,i}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.20}$$

where $\mathbf{x}_{e,i}$ are the coordinates of the middle point of the edge e_i, again $i \equiv i \pmod 4$. We use superscript (2) to denote the order of the function space. We can restrict the edge basis function to edges as

$$\varphi^{(2)}_{e,i}|_{e,j} = \begin{cases} \dfrac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})}, & i = j, \\ 0, & i \neq j. \end{cases} \tag{1.21}$$

In Figure 1.4, we present edge basis function $\varphi^{(2)}_{e,0}$ on a general quadrilateral element, where $\varphi^{(2)}_{e_0}$ evaluates 0 on edges $\{e_i, \quad i \neq 0\}$, and evaluates to a quadratic function on edge e_0.

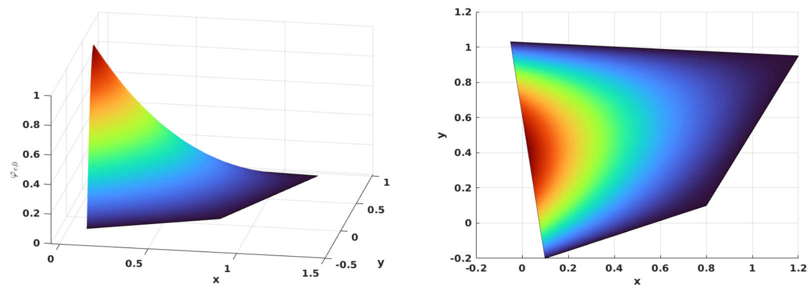


Figure 1.4: Edge nodal basis function $\varphi^{(2)}_{e,0}$ in $\mathcal{DS}_2$.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^{(2)}_{v,i}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1}\lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi^{2}_{e,i}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1}\lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi^{(2)}_{e,i}(\mathbf{x}_{v,i})}, \tag{1.22}$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i, and $i \equiv i \pmod 4$ as well. At the nodal points

$$\varphi^{(2)}_{v,i}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} = \mathbf{x}_{v,i}, \\ 0, & \mathbf{x} = \mathbf{x}_{v,j} \text{ or } \mathbf{x}_{e,k}, \quad j \neq i,\ \forall k. \end{cases} \tag{1.23}$$

In Figure 1.5, we show $\varphi^{(2)}_{v,0}$ on a general quadrilateral.

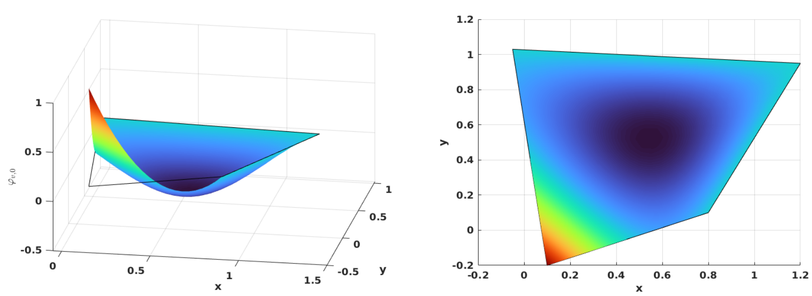


Figure 1.5: Vertex nodal basis function $\varphi^{2}_{v,0}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.3.2 First-order Space $\mathcal{DS}_1$

The dimension of the first-order polynomials in two dimension is also calculated by equation (1.9) as

$$\dim \mathbb{P}_1(\mathbb{R}^2) = \binom{1+2}{2} = \frac{(3)!}{2!1!} = 3, \tag{1.24}$$

while a quadrilateral element requires a minimum of 4 vertex dofs. Therefore, we supplement polynomial space $\mathbb{P}_1(E)$ with one additional function. The supplemental function for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order direct serendipity space as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \operatorname{span}\{R\}, \tag{1.25}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of the rational function R is not unique. We present one way to construct $R(\mathbf{x})$ using edge nodal basis function (1.20) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}), \tag{1.26}$$

where the auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{1.27}$$

In Figure 1.6, we plot the rational function $R(\mathbf{x})$ on a general quadrilateral.

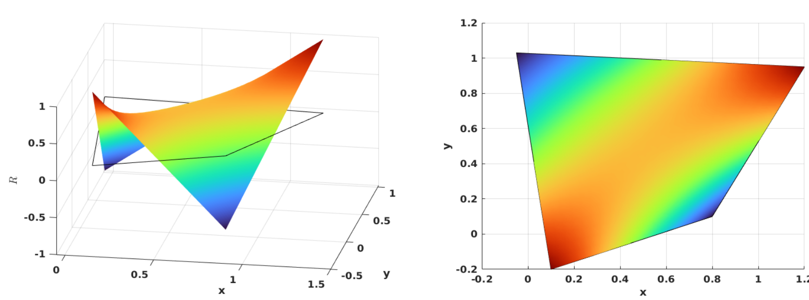


Figure 1.6: Supplemental rational function R in $\mathcal{DS}_1$.

We then define vertex nodal basis function accordingly as

$$\varphi_{v,i}^{(1)}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1+(-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{1.28}$$

where $m \equiv i+1 \pmod 2$, and $i \equiv i \pmod 4$. In Figure 1.7, we show the vertex nodal basis function $\varphi_{v,0}^{(1)}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertices.

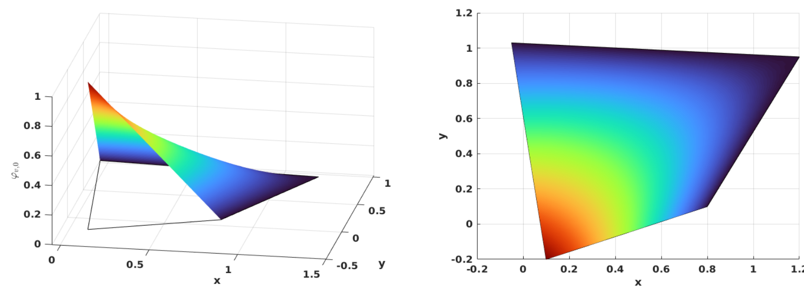


Figure 1.7: Vertex nodal basis function in $\mathcal{DS}_1$.

1.3.3 An Interpolation Operator

For completeness, we describe an interpolation operator mapping onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$, following Arbogast et al. (2022). For each node a_i, the associated K_i has different meanings depending on the type of the nodal basis functions. If the basis function is associated with an edge node, we set $K_i = e_i$, where e_i is the corresponding edge. For the basis function associated with a vertex node, we set K_i to be any edge e containing node a_i, with the restriction that if $a_i \in \partial\Omega$, then $e \in \partial\Omega$.

An interpolation operator $\mathcal{I}_h^r : W_p^l(\Omega) \to \mathcal{DS}_r$ is defined as

$$\mathcal{I}_h^r v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y})\, d\mathbf{y} \in \mathcal{DS}_r, \tag{1.29}$$

where $1 \le p \le \infty$ and $l > 1/p$ (but $l \ge 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x})\varphi_j(\mathbf{x})\, d\mathbf{x} = \delta_{i,j}, \quad i,j \in \{0,1,2,\ldots,N_r\}, \tag{1.30}$$

where N_r is the number of nodal basis functions. We present the following two lemmas regarding the constructed interpolation operator.

Lemma 1.3. *(Boundedness of the interpolation operator.)*

Let $\mathfrak{T}_h$ be uniformly shape regular according to Definition 1.3 *with shape regularity parameter σ_*. Let $v \in W_p^l(\Omega)$, with $1 \le p \le \infty$ and $l > q/p$. Moreover, assume that $R_{i,i+2}$ is a uniformly differentiable functions of the vertices of E up to order m. Then for $r = 1, 2$, $E \in \mathfrak{T}_h$, $1 \le q \le \infty$, and any non negative integer m,*

$$\|\mathfrak{I}_h^r v\|_{W_q^m(E)} \le C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \tag{1.31}$$

where $E^* = \cup_{F \in \mathfrak{T}_h, F \cap E \neq \emptyset} F$ and $|\cdot|_{W_p^k}$ is the seminorm of k-th order derivatives.

Proof. See Arbogast et al. (2022). □

Combining Lemma 1.3 and the Bramble-Hilbert lemma 1.2, we obtain local and global error estimation.

Lemma 1.4. *(Approximability of the interpolation operator.)*

Under the assumption of Lemma 1.3, *there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \le p \le \infty$ and $l > 1/p$ or ($l \ge 1$ if $p = 1$),*

$$\|v - \mathfrak{I}_h^r v\|_{W_p^m(E)} \le C h_E^{l-m} |v|_{W_p^l(E^*)}, \quad 0 \le m \le \min(l, r+1). \tag{1.32}$$

Moreover, there exists a constant $C = C(r, \sigma_*) > 0$, independent of $h = \max_{E \in \mathfrak{T}} h_E$, such that for all functions $v \in W_p^l(\Omega)$, $p < \infty$

$$\left(\sum_{E \in \mathfrak{T}_h} \|v - \mathfrak{I}_h^r v\|_{W_p^m(E)}^p \right)^{1/p} \le C h^{l-m} |v|_{W_p^l(\Omega)}, \quad 0 \le m \le \min(l, r+1). \tag{1.33}$$

Proof. See Arbogast et al. (2022). □

1.4 Mixed Finite Elements

In this section, we describe our mixed finite element approximation for the scaled coupled system (**??**) to (**??**). We begin by presenting the mixed variational formulations of the Stokes and Darcy model problems separately. Subsequently, we introduce H^1- and $H(\mathrm{div})$-conforming mixed finite element spaces. We describe the construction of mixed finite elements from the aforementioned direct serendipity space. Recall Ciarlet's definition Ciarlet (2002) of a finite element.

Definition 1.4. A triple $(E, X(E), \{\phi_1, \ldots, \phi_n\})$ defines a *finite element*, where

- The element $E \in \mathcal{T}_h$;
- The space of shape functions is $X(E)$ with basis $\{\varphi_1, \ldots, \varphi_n\}$,where the dimension of the shape function space is $\dim(X(E)) = n$;
- A set of continuous and linear functionals is $\{\phi_1, \ldots, \phi_n\}$, defined on a subset $\mathcal{X}(E)$ of the energy space containing the finite element space $X(E)$ as

$$\phi_j : \mathcal{X}(E) \to \mathbb{R}, \tag{1.34}$$

such that the unisolvence condition is achieved as

$$\langle \phi_j, \varphi_i \rangle = \delta_{ij}, \quad i, j \in \{1, 2, 3, \ldots, n\}. \tag{1.35}$$

In later subsections, we give an explicit construction of the nodal basis functions and degrees of freedom (values of the $\phi_j,\ j = 1, 2, \ldots, n$) that is associated to them, thereby completing the definition of the finite element.

1.4.1 A Darcy Model Problem

Consider a model problem for Darcy flow equipped with homogeneous boundary condition as

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{f}, \text{ in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0, \text{ in } \Omega, \\ \mathbf{u} \cdot \boldsymbol{\nu} &= 0, \text{ on } \partial\Omega. \end{aligned} \tag{1.36}$$

We seek the displacement and pressure solution pair $(\mathbf{u}, p)$ in the identical test and trial spaces

$$\begin{aligned}
\mathbb{V}_D &= \{\mathbf{v} \in (H(\mathrm{div};\Omega))^2 : \mathbf{v}\cdot\boldsymbol{\nu} = \mathbf{0} \;\text{ on }\; \partial\Omega\},\\
\mathbb{U}_D &= \{\mathbf{u} \in (H(\mathrm{div};\Omega))^2 : \mathbf{u}\cdot\boldsymbol{\nu} = \mathbf{0} \;\text{ on }\; \partial\Omega\} = \mathbb{V}_D,\\
\mathbb{W}_D &= \left\{p \in L^2(\Omega) : \int_\Omega p\,d\mathbf{x} = 0\right\} = L^2(\Omega)/\mathbb{R}.
\end{aligned} \tag{1.37}$$

The standard mixed weak form reads as follows:

Find $\mathbf{u} \in \mathbb{U}_D$ and $p \in \mathbb{W}_D$ such that

$$\begin{aligned}
(\mathbf{u}, \mathbf{v}) - (p, \nabla\cdot\mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}_D,\\
(\nabla\cdot\mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}_D.
\end{aligned} \tag{1.38}$$

We introduce two finite-dimensional subspaces $\mathbb{V}_{D,h} \subset \mathbb{V}_D$ and $\mathbb{W}_{D,h} \subset \mathbb{W}_D$, and consider the discretized problem:

Find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ such that

$$\begin{aligned}
(\mathbf{u}_h, \mathbf{v}_h) - (p_h, \nabla\cdot\mathbf{v}_h) &= (\mathbf{f}, \mathbf{v}_h), \quad \forall \mathbf{v}_h \in \mathbb{V}_{D,h},\\
(\nabla\cdot\mathbf{u}_h, \omega_h) &= 0, \qquad\quad \forall \omega_h \in \mathbb{W}_{D,h}.
\end{aligned} \tag{1.39}$$

In the next subsection, we present a construction of the discretized space pair $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ for this Darcy model problem.

1.4.2 A reduced $H(\mathbf{div})$ conforming space $\mathbb{V}_1^0$

We have some choice to discretize the space $\mathbb{V}_D$. For example, we can choose the classical Ravier-Thomas element Raviart and Thomas (1977). The serendipity space is the natural precursor of the Brezzi-Douglas-Marini element Brezzi et al. (1985) related by a de Rham complex. We generate a reduced first-order $H(\mathrm{div})$ approximation from the direct serendipity space, where $\nabla\cdot\mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$. We can also recover the Arbogast-Correa element Arbogast and Correa (2016) if we use the mapped serendipity space, as defined in that paper.

We write the de Rham exact sequence for the related energy spaces as

$$
\begin{array}{ccccccc}
\mathbb{R} \overset{\mathrm{i}}{\hookrightarrow} H^1 & \xrightarrow{\text{curl}} & H(\mathrm{div}) & \xrightarrow{\text{div}} & L^2 & \xrightarrow{0} & 0 \\
\cup & & \cup & & \cup & & \\
\mathcal{DS}_2 & \xrightarrow{\text{curl}} & \mathbf{V}_1^0 & \xrightarrow{\text{div}} & \mathbb{P}_0 & & .
\end{array}
\tag{1.40}
$$

We define the reduced first-order $H(\mathrm{div})$ conforming space as the first-order polynomial space supplemented with the curl of the second-order direct serendipity supplemental space of $\mathcal{DS}_2$, i.e.,

$$\mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \operatorname{curl} \mathbb{S}_2^{\mathcal{DS}}(E). \tag{1.41}$$

Here, $\mathbb{V}_1^0(E)$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements.

The dimension of this reduced first-order $H(\mathrm{div})$ conforming space is

$$\dim(\mathbb{V}_1^0(E)) = 2 \times 3 + 2 = 8. \tag{1.42}$$

On each edge of a quadrilateral element E, two independent basis functions are defined: one with vanishing average normal flux across the edge, and the other with non-vanishing average flux.

We state the degrees of freedom as normal fluxes on each edge as

$$DOF_{e_i}(\lambda_1) = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, \lambda_1 \rangle_{e_i}, \quad \lambda_1 \in \mathbb{P}_1, \tag{1.43}$$

where λ_1 is a linear function restricted to edge e_i, and $\dim(\mathbb{P}_1) = 2$. We then present a construction of two independent shape functions on each edge: $\boldsymbol{\varphi}_{l,i}$ and $\boldsymbol{\varphi}_{c,i}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \tag{1.44}$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{1.45}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i \, ds = 0$.

To begin with, we take the curl of the auxiliary rational function R_i (1.17) as

$$\operatorname{curl} R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{1.46}$$

where $i \equiv i \pmod 4$. We present one way to construct the "linear" shape function $\boldsymbol{\varphi}_{l,i}$ in two steps as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \operatorname{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\operatorname{curl}R_i. \tag{1.47}$$

We then normalize it such that l_i varies from -1 to 1 when restricted to edge e_i as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.48}$$

where the index $i \equiv i \pmod 4$, and the length of edge e_i is $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 1.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing an edge, and show the normal flux, but not the tangential flux joining continuously as a linear function on the shared edge.

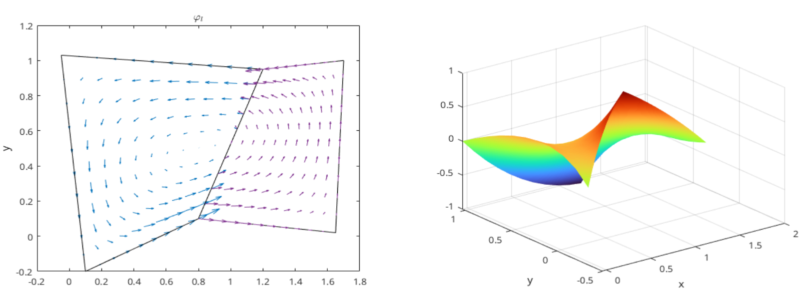


Figure 1.8: "Linear" shape function $\boldsymbol{\varphi}_l$.

The construction of the constant basis functions is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal function $\phi^*_{v,i} \in \mathcal{DS}_2(E)$ such that $\phi^*_{v,i}(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi^*_{v,i}(\mathbf{x}) = \varphi^2_{v,i}(\mathbf{x}) + \frac{1}{2}\big[\varphi^2_{e,i} + \varphi^2_{e,i+1}\big], \tag{1.49}$$

where $i \equiv i \pmod 4$. We then define $\boldsymbol{\varphi}^*_{v,i}(\mathbf{x}) = \operatorname{curl} \phi^*_{v,i}$ so that

$$\boldsymbol{\varphi}^*_{v,i} \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i+1, \\ 0, & j = i+2, i+3. \end{cases} \tag{1.50}$$

The "constant" basis function on the edge can now be defined as

$$\boldsymbol{\varphi}_{c,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|\boldsymbol{\varphi}^*_{v,i} + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \tag{1.51}$$

where $i \equiv i \pmod 4$. The "constant" shape function $\boldsymbol{\varphi}_{c,i}$ has a normal flux of constant value 1 on edge e_i and evaluates to 0 normal flux on all the other edges. In Figure 1.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing an edge, and show that the normal flux joins continuously as a constant 1 on the shared edge.

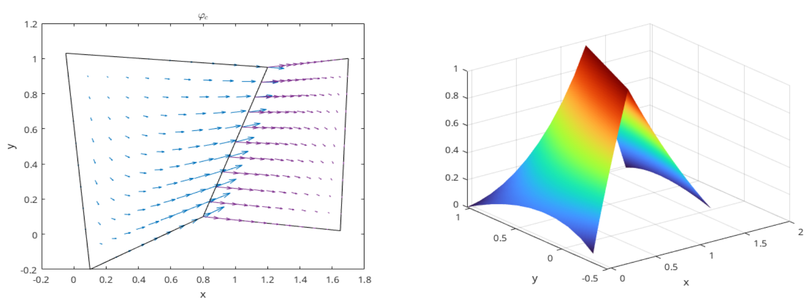


Figure 1.9: "Constant" shape function $\boldsymbol{\varphi}_c$.

1.4.3 Stability and Approximation Properties for $\mathbb{V}_1^0$

The accuracy and stability of $\mathbb{V}_1^0$ has been well studied and tested in the work by Arbogast et al. (2022). We state the result directly for completeness. They define a projection operation π that preserves element average divergence and average edge normal fluxes,

$$\pi : H(\mathrm{div}; \Omega) \cap (L^{2+\epsilon}(\Omega))^2 \to \mathbb{V}_1^0, \tag{1.52}$$

where $\epsilon > 0$. This global projection operator π is a combination of local operators π_E defined on each element E according to the degrees of freedom. The operator π

also satisfies the commuting diagram property

$$\mathcal{P}_{W_0} \nabla \cdot \mathbf{v} = \nabla \cdot \pi \mathbf{v}, \tag{1.53}$$

where $\mathcal{P}_{W_0}$ is the L^2-orthogonal projection operator onto $W_0 = \nabla \cdot \mathbb{V}_1^0$, which is a space of piecewise constant values.

Lemma 1.5. *Stability and approximation properties for $\mathbb{V}_1^0$.*

Under the assumptions of Lemma 1.3 and Lemma 1.2, there exists a constant $C > 0$, independent of $\mathcal{T}_h$ and $h > 0$, such that

$$\begin{aligned}
\|\mathbf{u} - \pi_D \mathbf{u}\|_{0,\Omega} &\leq C \|\mathbf{u}\|_{k,\Omega} h^k, && k = 1, 2, \\
\|\nabla \cdot (\mathbf{u} - \pi_D \mathbf{u})\|_{0,\Omega} &\leq C \|\nabla \cdot \mathbf{u}\|_{k,\Omega} h^k, && k = 0, 1, \\
\|p - \mathcal{P}_{W_s} p\|_{0,\Omega} &\leq C \|p\|_{k,\Omega} h^k, && k = 0, 1.
\end{aligned} \tag{1.54}$$

Moreover, the discrete inf-sup condition holds

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^0} \frac{(\omega_h, \nabla \cdot \mathbf{v}_h)}{\|\mathbf{v}_h\|_{H(div)}} \geq \gamma \|\omega_h\|_{0,\Omega}, \quad \forall \omega_h \in W_0, \tag{1.55}$$

for some $\gamma > 0$ independent of $\mathcal{T}_h$ and $h > 0$.

Proof. See Arbogast et al. (2022). □

1.4.4 A Stokes Model Problem

Consider a model problem of an incompressible isotropic equation equipped with homogeneous boundary condition as

$$\begin{aligned}
-\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\
\nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\
\mathbf{u} &= \mathbf{0} \quad \text{on } \partial\Omega.
\end{aligned} \tag{1.56}$$

We choose energy spaces for the displacement-pressure pair $(\mathbf{u}, p)$ in the identical test and trial spaces as

$$\begin{aligned}
\mathbb{V}_S &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \partial\Omega\} = (H_0^1(\Omega))^2, \\
\mathbb{U}_S &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{0} \text{ on } \partial\Omega\} = \mathbb{V}_S, \\
\mathbb{W}_S &= \left\{p \in L^2(\Omega) : \int_\Omega p \, d\mathbf{x} = 0\right\} = L^2(\Omega)/\mathbb{R}.
\end{aligned} \tag{1.57}$$

We consider the standard mixed variational form as

Find $\mathbf{u} \in \mathbb{U}_S$ and $p \in \mathbb{W}_S$ such that

$$\begin{aligned} (\nabla\mathbf{u}, \nabla\mathbf{v}) - (p, \nabla\cdot\mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}_S, \\ (\nabla\cdot\mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}_S. \end{aligned} \tag{1.58}$$

We introduce two finite-dimensional subspaces $\mathbb{V}_{S,h} \subset \mathbb{V}$ and $\mathbb{W}_{S,h} \subset \mathbb{W}$, and consider the discretized problem:

Find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ such that

$$\begin{aligned} (\nabla\mathbf{u}_h, \nabla\mathbf{v}_h) - (p_h, \nabla\cdot\mathbf{v}_h) &= (\mathbf{f}, \mathbf{v}_h), \quad \forall \mathbf{v}_h \in \mathbb{V}_{S,h}, \\ (\nabla\cdot\mathbf{u}_h, \omega_h) &= 0, \qquad\quad \forall \omega_h \in \mathbb{W}_{S,h}. \end{aligned} \tag{1.59}$$

Now we construct a pair of discretized space $\mathbb{V}_{S,h}\times\mathbb{W}_{S,h}$ for this Stokes model problem in the next subsection.

1.4.5 Modified Bernardi-Raugel Element

Stable mixed finite element pairs for the Stokes problem have been widely and thoroughly studied on triangles and rectangles. One notable construction on rectangles was discussed in Bernardi and Raugel (1985) and Fortin (1981), where polynomial spaces are supplemented with bubble functions. Consider the standard type of Bernardi-Raugel (BR) element using the space of tensor product polynomials

$$(\mathbb{Q}_{1,1}(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \subset (\mathbb{Q}_{2,2}(E))^2, \tag{1.60}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined for each edge e_i. A similar but more general type of element was discussed by Arbogast and Wheeler (2005).

We now modify the original BR element to extend it to a new finite element suited to quadrilaterals using the direct serendipity space as

$$\mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \subset (\mathcal{DS}_2(E))^2, \tag{1.61}$$

where $\dim \mathbb{V}_1^{\mathcal{BR}}(E) = 12$. One natural choice for the supplemental bubble function $\mathbf{b}_i$ is the previously defined edge nodal basis function $\varphi_{e,i}^{(2)} \in \mathcal{DS}_2$ times the unit normal, i.e.,

$$\mathbf{b}_i(\mathbf{x}) = \varphi_{e,i}^{(2)}(\mathbf{x})\boldsymbol{\nu}_i, \tag{1.62}$$

which is quadratic when restricted to edge e_i according to (1.21). The quadratic bubble is guaranteed to be conforming since it evaluates to 0 on two vertices and 1 at the middle point of the edge due to normalization, which is uniquely determined by these three points. The integral of this quadratic bubble can be accurately computed by a three-point Gaussian quadrature rule.

We state the degrees of freedom of this type of element as follows.

Definition 1.5. For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following 12 degrees of freedom:

(i) for corner point v_i and Cartesian direction unit normal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF_{(v_i,\mathbf{i})}^{(\mathrm{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF_{(v_i,\mathbf{j})}^{(\mathrm{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) for edge e_i,

$$DOF_{e_i}^{(ii)} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle.$$

We give an explicit construction of the shape functions. We choose the nodal basis function on vertex v_i in two Cartesian directions as

$$N_{v_i,x} = \begin{bmatrix} \varphi_{v,i}^{(1)}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \varphi_{v,i}^{(1)}(\mathbf{x}) \end{bmatrix}, \tag{1.63}$$

where $\varphi_{v,i}^{(1)}$ is the vertex nodal shape function in $\mathcal{DS}_1(E)$. We choose the edge basis function on edge e_i as (1.62), i.e.,

$$N_e = \varphi_{e,i}^{(2)}\boldsymbol{\nu}_i \tag{1.64}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i. Therefore the bubble functions cannot be in the same space as the nodal basis functions. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function.

We pair it with the space

$$\mathbb{W}_0 = \{\omega \in L^2(\Omega); \quad \omega|_E \in P_0\}/\mathbb{R}, \tag{1.65}$$

which has piecewise constant values for each element E. We provide stability and approximation analysis for this space pair $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ in the next subsection.

1.4.6 Stability and Approximation Properties for $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$

Before analyzing the stability and approximation properties for the new modified BR element, we state one useful lemma.

Lemma 1.6. *Let E be a nondengenate convex quadrilateral element that is shape regular as in definition* 1.1. *For each integer $m \geq 0$ and real number $p \in [1, \infty]$ there exist positive constants C_1 and C_2, possibly depending on σ^* but otherwise independent of the geometry of E, such that the following upper bounds hold for all $v \in W_p^m(E)$:*

$$\begin{cases} |v|_{W_p^0(E)} \leq C_1 h_E^{2/p} |\tilde{v}|_{W_p^0(\tilde{E})}, \\ |v|_{W_p^1(E)} \leq C_2(\sigma^*, p) h^{2/p-1} |\tilde{v}|_{W_p^1(\tilde{E})}, \end{cases} \tag{1.66}$$

where $\tilde{E}$ represents the local reference quadrilateral element $\tilde{E}$ illustrated in Figure 1.1, *and $\tilde{v}$ is the function defined on the local reference element $\tilde{E}$.*

Proof. See Girault and Raviart (2011). □

We reuse the interpolation operator $\mathcal{I}_h^1$ defined in equation (1.29). Define the projection operator $\pi \in (H_0^1(\Omega))^2 \to \mathbb{V}_1^{\mathcal{BR}} \cap (H_0^1(\Omega))^2$ as a combination of local projection operators π_E as:

(i) For each vertex $v \in \partial E$

$$\pi_E \mathbf{v}(\mathbf{x}_v) = \mathcal{I}_h^1 \mathbf{v}(\mathbf{x}_v); \tag{1.67}$$

(ii) For each edge $e \subset \partial E$,

$$\int_e (\pi_E \mathbf{v} - \mathbf{v}) \cdot \boldsymbol{\nu} \, ds = 0. \tag{1.68}$$

Lemma 1.7. *For all* $\mathbf{v} \in (H^k(\Omega))^2 \cap (H_0^1(\Omega))^2, \quad k = 1, 2$, *the operator* π_E *defined by* (1.67)–(1.68) *satisfies*

$$(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0, \quad \forall q_h \in \mathbb{W}_0 \tag{1.69}$$

Assuming shape regularity as defined in Definition 1.1*, the global projection operator* π *satisfies the error bound*

$$\|\mathbf{v} - \pi \mathbf{v}\|_{H^1(\Omega)} \leq C h^m \|\mathbf{v}\|_{H^{m+1}(\Omega)}, \quad \forall \mathbf{v} \in (H^{m+1}(\Omega))^2, \ m = 0, 1. \tag{1.70}$$

Proof. According to the divergence theorem, we have

$$\int_E \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v}) \, d\mathbf{x} = \sum_{i=0}^{3} \int_{e_i} (\mathbf{v} - \pi_E \mathbf{v}) \cdot \boldsymbol{\nu} \, ds = 0. \tag{1.71}$$

Given that q_h is piecewise constant on the element E, it is clear that we have

$$(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0, \quad \forall q_h \in \mathbb{W}_0. \tag{1.72}$$

From the definition of the degrees of freedom (1.67) and (1.68), we decompose the projection operator π_E as

$$\pi_E \mathbf{v} = \mathcal{I}_h^1 \mathbf{v} + \sum_{i=0}^{3} \alpha_i \mathbf{b}_i, \tag{1.73}$$

where the coefficient

$$\alpha_i = \frac{\int_{e_i} (\mathbf{v} - \mathcal{I}_h^1 \mathbf{v}) \cdot \boldsymbol{\nu}_i \, ds}{\int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i \, ds}. \tag{1.74}$$

The approximation property of the interpolation operator $\mathcal{I}_h^1$ has been discussed in Lemma 1.4,

$$\left\|\mathbf{v} - \mathcal{I}_h^1 \mathbf{v}\right\|_{H^m(E)} \leq h^{n-m} |\mathbf{v}|_{H^n(E^*)}, \quad m = 0, 1, \quad n = 1, 2. \tag{1.75}$$

where $E^* = \cup_{F\in\mathfrak{T}_h, F\cap E\neq\emptyset}$ defined in Lemma 1.3.

Lemma 1.6 indicates an estimation of the bubble function as

$$|\mathbf{b}_i|_{H^m(E)} \leq C(\sigma^*)h^{1-m}|\tilde{\mathbf{b}}_i|_{H^1(\tilde{E})} \leq C(\sigma^*)h^{1-m}, \quad m = 0, 1. \tag{1.76}$$

We continue to estimate $|\alpha_i|$. Note

$$\int_{e,i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i \, ds = \int_{e,i} \varphi_{e,i}^{(2)} \, ds = |e_i| \int_{\tilde{e}_i} \tilde{\varphi}_{e,i}^{(2)} \, d\tilde{s} = C_1|e_i|. \tag{1.77}$$

Then

$$\begin{aligned} |\alpha_i| = \left| \frac{\int_{e_i} (\mathbf{v} - \mathfrak{I}_h^1\mathbf{v}) \cdot \boldsymbol{\nu}_i \, ds}{\int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i \, ds} \right| &\leq C \int_{\tilde{e},i} |\tilde{\mathbf{v}} - \tilde{\mathfrak{I}}_h^1 \tilde{\mathbf{v}}| \, ds \\ &\leq C \left\| \tilde{\mathbf{v}} - \tilde{\mathfrak{I}}_h^1 \mathbf{v} \right\|_{L^2(\tilde{e}_i)}. \end{aligned} \tag{1.78}$$

As the trace mapping is contimuous from $H^1(\tilde{E})$ into $L^2(\tilde{e})$, according to the Trace Theorem 1.1,

$$\begin{aligned} |\alpha_i| &\leq C \left\| \tilde{\mathbf{v}} - \tilde{\mathfrak{I}}_h^1 \mathbf{v} \right\|_{H^1(\tilde{E})} \\ &\leq C \Big(\left\| \tilde{\mathbf{v}} - \tilde{\mathfrak{I}}_h^1 \mathbf{v} \right\|_{L^2(E)} + h^{-1} \left| \tilde{\mathbf{v}} - \tilde{\mathfrak{I}}_h^1 \mathbf{v} \right|_{H^1(E)} \Big). \end{aligned} \tag{1.79}$$

According to Lemma 1.4,

$$|\alpha_i| \leq Ch^{k-1} \|\mathbf{v}\|_{H^k(E^*)}, \quad k = 1, 2. \tag{1.80}$$

The overall estimate is then

$$|\alpha_i||\mathbf{b}_i|_{H^m(E)} \leq Ch^{k-m} \|\mathbf{v}\|_{H^k(E^*)}, \quad m = 0, 1, \quad k = 1, 2. \tag{1.81}$$

Combining with the interpolation error estimate result in equation (1.75), we get a local error estimate as

$$\|\mathbf{v} - \pi_E \mathbf{v}\|_{H^1(E)} \leq Ch \|\mathbf{v}\|_{H^2(E)}. \tag{1.82}$$

The global error estimate is obtained by combining local results together. □

In addition to the approximability lemma 1.4, we also need a stability result.

Lemma 1.8. *Assuming that* $\mathbf{v} \in (H^1(\Omega))^2$*, the linear operator* π *is bounded on* $(H^1(\Omega))^2$ *independently of* h*, i.e.,*

$$\|\pi \mathbf{v}\|_{H^1(\Omega)} \leq C \|\mathbf{v}\|_{H^1(\Omega)} . \tag{1.83}$$

Proof. According to the triangle inequality,

$$\|\pi_E \mathbf{v}\|_{H^1(E)} \leq \|\mathbf{v}\|_{H^1(E)} + \|\mathbf{v} - \pi_E \mathbf{v}\|_{H^1(E)} \leq (1 + C) \|\mathbf{v}\|_{H^1(E)} , \tag{1.84}$$

where C comes from Lemma 1.7. The global stability result can be obtained by combining local results. $\square$

Theorem 1.9. *With the established approximability result stated in Lemma* 1.7*, and the stability result stated in Lemma* 1.8*, for a convex domain* Ω*, there exists a constant* $\gamma > 0$ *such that*

$$\inf_{q_h \in \mathbb{W}_0} \sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathfrak{BR}}} \frac{(\nabla \cdot \mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \gamma > 0. \tag{1.85}$$

Proof. Note that

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathfrak{BR}}} \frac{(\nabla \cdot \mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \frac{(\nabla \cdot \pi \mathbf{v}, q_h)}{\|\pi \mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}}. \tag{1.86}$$

Recall that the inf-sup condition holds for the continuous space as

$$\inf_{q \in L^2(\Omega)} \sup_{\mathbf{v} \in (H^1(\Omega))^2} \frac{(\nabla \cdot \mathbf{v}, q)}{\|\mathbf{v}\|_{H^1(\Omega)} \|q\|_{L^2(\Omega)}} \geq \gamma > 0. \tag{1.87}$$

With the condition $(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0$ proved in Lemma 1.7 we have

$$\frac{(\nabla \cdot \pi \mathbf{v}, q_h)}{\|\pi \mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} = \frac{(\nabla \cdot \mathbf{v}, q_h)}{\|\pi \mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \frac{\gamma}{C} > 0, \tag{1.88}$$

where C here is the constant in (1.83). $\square$

We conclude that our space $\mathbb{V}_1^{\mathfrak{BR}} \times \mathbb{W}_0$ forms a inf-sup stable pair with first order overall accuracy.

1.4.7 Numerical Test

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem but with a non-homogenous Dirichlet boundary condition, i.e., on the problem:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{g} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.89}$$

We assume that $\mathbf{g} \in (H^{1/2}(\partial\Omega))^2$ and understand $\tilde{\mathbf{g}} \in (H^1(\Omega))^2$ as a continuous extension from the boundary into the domain. We define a simple choice of energy spaces

$$\begin{aligned} \mathbb{V} &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \ \text{ on } \partial\Omega\}, \\ \mathbb{U} &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g} \ \text{ on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V}, \\ \mathbb{W} &= \{p \in L^2(\Omega)\}/\mathbb{R}. \end{aligned} \tag{1.90}$$

We write the problem in weak form as:

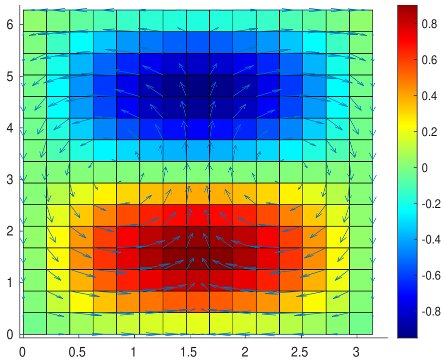
Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}. \end{aligned} \tag{1.91}$$

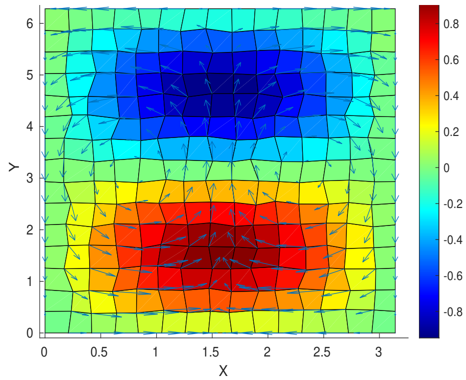
We consider the true solution of test problem (1.91) defined on a square region $\Omega = [0, \pi] \times [0, 2\pi]$. We manufacture a displacement and pressure as

$$\mathbf{u} = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \quad p = \sin(x)\sin(y), \tag{1.92}$$

which has 0 normal flux on the boundary. We deal with the constant kernal by enforcing $\int_\Omega p \, d\mathbf{x} = 0$ and solve the linear system directly. The numerical results are computed on rectangular and logically rectangular quadrilateral meshes as depicted in Figures 1.10a and 1.10b, respectively. We calculate the L^2 norm of the residual



(a) Rectangular Grids. Shown are pressure (color) and velocity (arrows).



(b) Distorted Grids. Shown are pressure (color) and velocity (arrows).

Figure 1.10: Numerical results on rectangular and quadrilateral mesh.

(i.e., the difference) of the true and approximate pressure, velocity, and derivative of velocity.

In Table 1.1 and Table 1.2 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. A second order convergence rate is achieved by velocity. Pressure achieves some superconvergence due to the use of the mid point integration rule. We see second order convergence on the rectangular grid, and some rate better than the first on distorted grid. A first order convergence rate is achieved by derivative of velocity. These meet our expectations for optimal convergence order.

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.502E−1	1.052E−1	2.513E−2	6.130E−3
$L^2(\text{res}_{du})$	1.039E+0	4.463E−1	2.093E−1	1.019E−1
$L^2(\text{res}_p)$	2.632E+0	5.845E−1	1.374E−1	3.330E−2

Table 1.1: Rectangular Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.603E−1	1.074E−2	2.567E−2	6.266E−3
$L^2(\text{res}_{du})$	1.070E+0	4.670E−1	2.207E−1	1.078E−1
$L^2(\text{res}_p)$	2.685E+0	5.981E−1	1.525E−1	4.580E−2

Table 1.2: Distorted Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

1.5 Coupled Degenerate Darcy-Stokes System

Now we consider a scaled nondimensionalized degenerate Darcy-Stokes coupled boundary-value problem:

Find two velocity-pressure potential pairs $(\check{\mathbf{v}}_r, \check{q}_l)$ and $(\check{\mathbf{v}}_s, \check{q})$ such that

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1.93}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.94}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{1.95}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.96}$$

and

$$\begin{aligned} \check{\mathbf{v}}_r &= \mathbf{g}_r \quad \text{on } \Gamma_{u,D}, \\ \check{\mathbf{v}}_s &= \mathbf{g}_s \quad \text{on } \Gamma_{u,S}, \\ p &= p_0 \quad \text{on } \Gamma_{i,D}, \\ \boldsymbol{\tau}\cdot\boldsymbol{\nu} - p &= \boldsymbol{t}_0 \quad \text{on } \Gamma_{i,S}, \end{aligned} \tag{1.97}$$

where $\Gamma_{u,D}$ and $\Gamma_{u,S}$ denote the part of boundary we assign essential boundary conditions for Darcy and Stokes problem respectively, and $\Gamma_{i,D}$ and $\Gamma_{i,S}$ represent the part of boundary we assign natural boundary conditions respectively. We have $\Gamma_{u,D}\cup\Gamma_{i,D} = \partial\Omega$ and $\Gamma_{u,S}\cup\Gamma_{i,S} = \partial\Omega$.

We expand the gradient term in equation (1.93) and the divergence term in equation (1.94) as

$$\begin{aligned} \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) &= \phi^{1/2+\Theta}\nabla\check{q}_l + \phi^{\Theta-1/2}\check{q}_l\nabla\phi, \\ \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) &= \phi^{1/2+\Theta}\nabla\cdot\check{\mathbf{v}}_r + \phi^{\Theta-1/2}\nabla\phi\cdot\check{\mathbf{v}}_r, \end{aligned} \tag{1.98}$$

where they are well-defined provided that

$$\phi^{\Theta-1/2}\nabla\phi \in (L^\infty(\Omega))^2. \tag{1.99}$$

A scaled Hilbert space is defined as follows for this specific scaled formulation, where the scaled velocity $\check{\mathbf{v}}_r$ should lie in.

Definition 1.6. Consider a scaled Hilbert space:

$$\mathbb{V}_D^\phi = H_\phi(\text{div};\Omega) = \Big\{\mathbf{v} \in (L^2(\Omega))^2 : \phi^{-1/2}\nabla\cdot(\phi^{1+\Phi}\mathbf{v}) \in L^2(\Omega)\Big\}, \tag{1.100}$$

which is a Hilbert space equipped with the inner product

$$(\mathbf{u},\mathbf{v})_{\mathbb{V}_D^\phi} = (\mathbf{u},\mathbf{v}) + (\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{u}), \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})). \tag{1.101}$$

The normal trace on $\partial\Omega$ is well-defined as

$$\left\|\phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu}\right\|_{H^{-1/2}(\partial\Omega)} < C_\Omega\Big\{\left\|\mathbf{v}\right\|_{L^2(\Omega)} + \left\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})\right\|_{L^2(\Omega)}\Big\} \tag{1.102}$$

subsequently, we have the space $H_\phi^{-1/2}(\text{div};\partial\Omega)$ understood as the image of this normal trace operator.

1.5.1 The weak formulation

We define the following function spaces for the scaled Darcy-Stokes coupled system as

$$\begin{aligned}
\mathbb{V}_D^\phi &= \Big\{\mathbf{v} \in H_\phi(\text{div};\Omega) : \phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu} = \mathbf{0} \text{ on } \Gamma_{u,D}\Big\},\\
\mathbb{U}_D^\phi &= \Big\{\mathbf{u} \in H_\phi(\text{div};\Omega) : \phi^{1/2+\Theta}\mathbf{u}\cdot\boldsymbol{\nu} = \mathbf{g}_r \text{ on } \Gamma_{u,D}\Big\} = \tilde{\mathbf{g}}_r + \mathbb{V}_D^\phi,\\
\mathbb{W}_D &= L^2(\Omega),\\
\mathbb{V}_S &= \Big\{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S}\Big\},\\
\mathbb{U}_S &= \Big\{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g}_s \text{ on } \Gamma_{u,S}\Big\} = \tilde{\mathbf{g}}_s + \mathbb{V}_S,\\
\mathbb{W}_S &= L^2(\Omega),
\end{aligned} \tag{1.103}$$

where each space is equipped with their natural norms. Here, $\tilde{\mathbf{g}}_r$ and $\tilde{\mathbf{g}}_s$ are finite energy lift of Dirichlet data $\mathbf{g}_r$ and $\mathbf{g}_s$ respectively.

We understand the boundary values in the sense of Trace Theorem 1.1 and the scaled normal trace operator (1.102). We assume regularity for Dirichlet data $\mathbf{g}_r \in (H_\phi^{1/2}(\mathrm{div};\Gamma_{u,D}))^2$ and $\mathbf{g}_s \in (H^{1/2}(\Gamma_{u,S}))^2$. We understand $\tilde{\mathbf{g}}_r \in H_\phi(\mathrm{div};\Omega)$ and $\tilde{\mathbf{g}}_s \in (H^1(\Omega))^2$ as a continuous extension from the boundary into the domain.

A compatibility condition enforcing mass conservation on the entire domain is needed if essential boundary condition is prescribed on the entire boundary $\partial\Omega$, i.e., when $\Gamma_{u,D} = \Gamma_{u,S} = \partial\Omega$ the Dirichlet data satisfies

$$\int_{\partial\Omega} (\mathbf{g}_r + \mathbf{g}_s)\cdot \boldsymbol{\nu}\, ds = 0. \tag{1.104}$$

The standard mixed form for the proposed coupled system reads as follows:

Find $\check{\mathbf{v}}_r \in \mathbb{U}_D^\phi$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{U}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) - (\check{q}, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall \boldsymbol{\psi}_s \in \mathbb{V}_S, \tag{1.105}$$

$$(\nabla\cdot\check{\mathbf{v}}_s, \omega) + \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega\Big) = 0 \qquad \forall \omega \in \mathbb{W}_S, \tag{1.106}$$

$$(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)) = 0, \qquad \forall \boldsymbol{\psi}_r \in \mathbb{V}_D^\phi, \tag{1.107}$$

$$(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \Big(\frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega_f\Big) = 0 \qquad \forall \omega_f \in \mathbb{W}_D. \tag{1.108}$$

We state well-posedness result of the scaled model as follows.

Lemma 1.10. *(Existence and Uniqueness)*

Assume the condition (1.99) *holds,* $0 \le \phi \le \phi^* < 1$. *There exists a unique solution to the scaled formulation* (1.105)–(1.108), *and it satisfies*

$$\|\check{\mathbf{v}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)\big\|_{(L^2(\Omega))^2} + \|\check{q}_l\|_{(L^2(\Omega))^2} + \|\check{\mathbf{v}}_s\|_{(H^1(\Omega))^2} + \|\check{q}\|_{(L^2(\Omega))^2}$$
$$\le C\big\{|\rho_r| + \|\tilde{\mathbf{g}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{g}}_r)\big\|_{(L^2(\Omega))^2} + \|\tilde{\mathbf{g}}_s\|_{H^1(\Omega)}\big\}. \tag{1.109}$$

Proof. See Arbogast et al. (2017). □

1.5.2 The scaled mixed finite element method

We discretize the scaled mixed formulation with two space pairs $\mathbb{V}_{D,h}\times\mathbb{W}_{D,h}\subset H(\mathrm{div};\Omega)\times L^2(\Omega)$ and $\mathbb{V}_{S,h}\times\mathbb{W}_{S,h}\subset(H^1(\Omega))^2\times L^2(\Omega)$. Recall the aforementioned inf-sup stable mixed finite element space pairs on $E\in\mathcal{T}_h$:

- For the Darcy part of the system.

$$
\begin{aligned}
\mathbb{V}_{D,h}(E) &= \mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \operatorname{curl}\mathbb{S}_2^{\mathcal{DS}}(E)\\
\mathbb{W}_{D,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E)
\end{aligned}
\tag{1.110}
$$

- For the Stokes part of the system.

$$
\begin{aligned}
\mathbb{V}_{S,h}(E) &= \mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0,\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3\}\\
\mathbb{W}_{S,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E).
\end{aligned}
\tag{1.111}
$$

We can impose essential boundary conditions by projecting extensions $\tilde{\mathbf{g}}_r$ and $\tilde{\mathbf{g}}_s$ into the finite element spaces as $\hat{\mathbf{g}}_r\in\mathbb{V}_{D,h}$ and $\hat{\mathbf{g}}_s\in\mathbb{V}_{S,h}$.

We also define

$$
\begin{aligned}
\mathbb{V}_{D,h,0} &= \{\mathbf{v}\in\mathbb{V}_{D,h} : \mathbf{v}\cdot\boldsymbol{\nu}=0 \ \text{ on } \ \Gamma_{u,D}\}\\
\mathbb{V}_{S,h,\mathbf{0}} &= \{\mathbf{v}\in\mathbb{V}_{S,h} : \mathbf{v}=\mathbf{0} \ \text{ on } \ \Gamma_{u,S}\}
\end{aligned}
\tag{1.112}
$$

The scaled mixed finite element method: Find $\check{\mathbf{v}}_{r,h}\in\mathbb{V}_{D,h,0}+\hat{\mathbf{g}}_r$, $\check{q}_{r,h}\in\mathbb{W}_{D,h}$, $\check{\mathbf{v}}_{s,h}\in\mathbf{V}_{S,h,\mathbf{0}}+\hat{\mathbf{g}}_s$ and $\check{q}_h\in\mathbb{W}_{S,h}$ such that

$$
(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}),\nabla\boldsymbol{\psi}_{s,h})-(\check{q}_h,\nabla\cdot\boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n},\boldsymbol{\psi}_{s,h}),\qquad \forall\boldsymbol{\psi}_{s,h}\in\mathbb{V}_{S,h,\mathbf{0}},\tag{1.113}
$$

$$
(\nabla\cdot\check{\mathbf{v}}_{s,h},\omega_h)-\Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_{l,h}-\phi^{1/2}\check{q}_h),\omega_h\Big)=0,\qquad \forall\omega_h\in\mathbb{W}_{S,h},\tag{1.114}
$$

$$
(\check{\mathbf{v}}_{r,h},\boldsymbol{\psi}_{r,h})-(\check{q}_{l,h},\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}))=0,\qquad \forall\boldsymbol{\psi}_{r,h}\in\mathbb{V}_{D,h,0},\tag{1.115}
$$

$$
(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}),\omega_{l,h})+\Big(\frac{1}{1-\phi}(\check{q}_{l,h}-\phi^{1/2}\check{q}_h),\omega_{l,h}\Big)=0,\qquad \forall\omega_{l,h}\in\mathbb{W}_{D,h}.\tag{1.116}
$$

1.5.3 Local Mass Conservation

Before we continue the discussion, we need to make sure that the formulation is well-defined, i.e., not dividing by zero. The term $\phi^{-1/2}$ in the symmetric equations (1.115) and (1.116) causes trouble when there is no melting, i.e., $\phi = 0$. In order to deal with it, we define an element-averaged variable ϕ_E as

$$\phi_E = \frac{1}{|E|} \int_E \phi(\mathbf{x})\, d\mathbf{x}, \tag{1.117}$$

where $|E|$ is the area of element E. Then the two terms involving divergence is modified as

$$\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \mathbf{v})|_E = \begin{cases} \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \mathbf{v}) & \text{if } \overline{\phi}_E \neq 0, \\ 0 & \text{if } \overline{\phi}_E = 0, \end{cases} \tag{1.118}$$

where the point-wise term $\phi^{-1/2}$ is modified to be a cell-averaged term $\phi_E^{-1/2}$. We expand the integral term involving divergence according to the divergence theorem as

$$\begin{aligned} \int_E \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h}) \omega_{l,h}\, d\mathbf{x} = & \int_{\partial E} \phi_E^{-1/2} \phi^{1+\Theta} \boldsymbol{\psi}_{r,h} \cdot \boldsymbol{\nu} \omega_{l,h}\, ds \\ & + \int_E \phi_E^{-1/2} \phi^{1+\Theta} \boldsymbol{\psi}_{r,h} \cdot \nabla \omega_{l,h}\, d\mathbf{x}. \end{aligned} \tag{1.119}$$

The second term vanishes when $\omega_{l,h}$ is constant on element E. We now propose our local mass conserved version by replacing the term $\phi^{1/2}$ in equation (1.114) with $\phi_E^{1/2}$ as

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}) \qquad \forall \boldsymbol{\psi}_{s,h}, \in \mathbb{V}_{S,h,\mathbf{0}}, \tag{1.120}$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega_h) - \Big(\frac{\phi \phi_E^{-1/2}}{1-\phi} (\check{q}_{l,h} - \phi_E^{1/2} \check{q}_h), \omega_h \Big) = 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}, \tag{1.121}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \tag{1.122}$$

$$(\phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \Big(\frac{\phi \phi_E^{-1}}{1-\phi} (\check{q}_{l,h} - \phi_E^{1/2} \check{q}_h), \omega_{l,h} \Big) = 0, \qquad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \tag{1.123}$$

To see local mass conservation of the fluid, let $E \in \mathcal{T}_h$ be any element. The equation (1.123) gives

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}) d\mathbf{x} + \int_E \frac{\phi}{1-\phi} (\check{q}_{l,h} - q_h) dx = 0, \tag{1.124}$$

and the equation (1.121) gives

$$\int_E \nabla \cdot \mathbf{v}_{s,h} d\mathbf{x} - \int_E \frac{\phi}{1-\phi} (q_{l,h} - q_h) dx = 0. \tag{1.125}$$

We define an auxiliary space ω_E being 1 on E and 0 elsewhere. We sum equations (1.121) and (1.123) together assuming that $\omega_h = \omega_E \in \mathbb{W}_{S,h}$ and $\omega_{l,h} = \phi_E^{-1/2} \omega_E \in \mathbb{W}_{D,h}$. The local mass conservation of the phase-averaged mixture is now retrieved as

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r + \check{\mathbf{v}}_s) d\mathbf{x} = 0. \tag{1.126}$$

1.5.4 Uniqueness and Convergence Properties

Lemma 1.11. *Assuming* (1.99) *holds, there exists a unique solution to the modified local mass conservation modified scaled mixed finite element method* (1.120)–(1.123).

Proof. See Arbogast et al. (2017). The local mass conservation modification replace the term $1/(1-\phi)$ with $(\phi\phi_E^{-1})/(1-\phi)$,which makes it completely analogous to the previous proof. Hence, we are not repeating here. □

We now derive a bound for the local mass conservation modified scaled mixed finite element method.

We derive a bound for the error by taking the difference of

We use the projection operators defined in

1.6 Saddle Point System

Saddle point system is widely spread in technical and science scenarios, such as constrained optimization problem, image reconstruction and optimal control. Saddle point system has been widely and systematically studied, e.g., by Benzi et al. (2005). In our case, saddle point system is formed along with the Mixed Finite Element discretization. In this section, we give a brief overview of properties of a saddle point system and some common ways to solve such system. An abstract saddle point system can be represented as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{g} \end{bmatrix}. \tag{1.127}$$

Assuming the following properties:

- $\mathbf{A}$ is symmetric: $\mathbf{A} = \mathbf{A}^T$;
- $\mathbf{A}$ is nonsingular: $\mathbf{A}^{-1}$ exists;
- $\mathbf{C}$ is symmetric and positive semidefinite;
- $\mathbf{B}$ is rank deficient.

1.6.1 Reduced Linear System

Now we discuss the saddle point system (SPS) induced by MFEM method in details including the boundary conditions. To begin with, we regroup dofs according to the boundary condition.

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_K & \mathbf{A}_M \\ \mathbf{A}_M^T & \mathbf{M} \end{bmatrix}, \tag{1.128}$$

where $\mathbf{A}_K$ consists of interaction within interior dofs and natural boundary condition dofs, and $\mathbf{A}_M$ consists of iteraction between interior dofs, natural boundary dofs and essential boundary dofs.

We perform the same treatment for matrix $\mathbf{B}$ and right hand side vector $\mathbf{f}$, by regrouping dofs according to its iteration with boundary conditions.

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_K \\ \mathbf{B}_M \end{bmatrix}, \text{and} \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_K \\ \mathbf{F}_M \end{bmatrix} \tag{1.129}$$

We rewrite the linear system with the regrouped dofs and form a reduced system as

$$\begin{bmatrix} \mathbf{A}_K & \mathbf{B}_K \\ \mathbf{B}_K^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_K - \mathbf{A}_M\mathbf{g} - \mathbf{g}_N \\ \mathbf{G} - \mathbf{B}_M^T\mathbf{g} \end{bmatrix}, \tag{1.130}$$

where $\mathbf{g}$ denotes the values prescribed for essential boundary, and $\mathbf{g}_N$ is the vector generated by the natural boundary condition part. The reformed reduced system is still a saddle point syste.

Now we take a closer look at the formed linear system. The no melting (zero porosity $\phi = 0$) region adds difficulties to solving the formed saddle point system. On one hand, the existence of zero porosity region guarantees that $\mathbf{B}$ matrix formed in Darcy system does not has full column rank. Therefore the schur complement

$$\mathbf{S} = -(\mathbf{C} + \mathbf{B}\mathbf{A}^{-1}\mathbf{B}^T),$$

is usually not invertable directly. On the other hand, in addition to the constant null space induced by gradient of pressure, the null space of $\mathbf{B}$ matrix $\ker(\mathbf{B})$ in our darcy problem is not fixed due to evloution of melting (change of zero porosity region).

Taking into consideration that the solver will be implemented in parallel, a iterative solver is more suitable than direct solver.

1.6.2 Schur Complement

To begin with, we discuss a "direct" way to solve this derived saddle point system. Now consider a general saddle point system (1.127) with symmetric positive-definite matrix $\mathbf{A}$ as

$$\begin{aligned} \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} &= \mathbf{f}, \\ \mathbf{B}^T\mathbf{x} - \mathbf{C}\mathbf{y} &= \mathbf{g}. \end{aligned} \tag{1.131}$$

We represent the solution vector $\mathbf{x}$ as

$$\mathbf{x} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{B}\mathbf{y}), \tag{1.132}$$

where $\mathbf{A}^{-1}$ is computed with conjugate gradient algorithm. We then eliminate $\mathbf{x}$ to represent $\mathbf{y}$ as

$$\mathbf{y} = \mathbf{S}^{-1}(\mathbf{g} - \mathbf{B}^T\mathbf{A}^{-1}\mathbf{f}), \tag{1.133}$$

where $\mathbf{S}^{-1}$ is computed with Minimal Resigual algorithm Paige and Saunders (1975) due to the formed Schur complement being a symmtric indefinite system.

1.6.3 Uzawa Iteration

Uzawa algorithm was originally introduced by K.J.Arrow et al. (1958), which gained popularity in computational fluid dynamics. It has later been applied to the case of discretizing Stokes problem with Mixed Finite Element. Some inexact inner solver to accelerate uzawa iteration was discussed by Elman and Golub (1994) and Bramble et al. (1997).

We start by stating an abstract inexact ezawa iteration algorithm in Algorithm 1. Some choose diagonal matrix $\alpha\mathbf{I}$ for precondictioner $\mathbf{Q}_A$. However, such

---

Algorithm 1 Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.

2: **while** res $>$ tol **do**

3: Obtain $\tilde{\mathbf{x}} = \mathbf{Q}_A^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$

4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$

5: Obtain $\tilde{\mathbf{y}} = \mathbf{Q}_B^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$

6: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

7: Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$

8: **end while**

---

simple matrix wouldn't generate a convergence result when applied to our coupled problem. In the next section, we introduce a modified algorithm that would solve our system.

1.6.4 Modified Uzawa Iteration

The linear system formed from the equations (1.120), (1.121), (1.122) and (1.123) is

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{1.134}$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{1.135}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{1.136}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{1.137}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system.

Algorithm 2 Modified Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.

2: **while** res $>$ tol **do**

3: Obtain $\tilde{\mathbf{x}} = \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$

4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$

5: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$

6: Obtain $\tilde{\mathbf{y}} = \mathrm{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$

7: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

8: Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$

9: **end while**

1.6.5 Numerical Test

We test our MFEM setup with three sets of problems. First, we test with a manufactured solution all prescribed with essential boundary condition, but the right hand side was modified. Second, we test a pseudo 1D problem with 2D elements with three different porosity scenarios. Third, we test a more complex yet a more realistic problem representing Mid Ocean Ridge.

Problem Setup 1: Manufactured Solution.

We solve a manufactured system of equations on the rectangular domain $(x, y) \in [-1, 1]^2$. We assume a constant porosity acorss the domain as ϕ. We compute the right hand side with prescribed functions as

$$\check{q} = 0 \qquad \check{q}_f = 2(x+y) \qquad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \qquad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \tag{1.138}$$

$$\nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f}(x+y) \tag{1.139}$$

$$\mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \qquad (\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I} = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix} \tag{1.140}$$

Substitute the compuated results into PDE

$$\begin{aligned}
\tilde{\mathbf{v}}_r + \phi_f^{1/2}\nabla\check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2}\begin{pmatrix} 2 \\ 2 \end{pmatrix} \\
\phi_f^{1/2}\nabla\cdot(\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}\check{q}_f &= \frac{-2}{1-\phi_f}(x+y) + \frac{1}{1-\phi_f}2(x+y) = 0 \\
\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}\check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) = 0 \\
-2(1-\phi_f)\nabla\cdot\sigma(\check{\mathbf{v}}_s) &= -2(1-\phi_f)\nabla\cdot(\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3}(\nabla\cdot\check{\mathbf{v}}_s)\mathrm{I}) = -\frac{8\phi_f^{1/2}}{3}\begin{pmatrix} 1 \\ 1 \end{pmatrix}
\end{aligned} \tag{1.141}$$

Problem Setup 2: Column Compaction.

In this problemsetup, we perform convergence test on a 2D grid with a 1D problem. We study a compacting column in one direction. The column extends over $y \in [-L, L]$ and has no flow through top and bottom boundaries. We prescribe the following boundary conditions

- Bottom Edge;
  - Stokes velocity $\check{\mathbf{v}}_s = \mathbf{0}$;
  - Darcy velocity $\check{\mathbf{v}}_l = \mathbf{0}$;
  - Normal flux is also determined to be zero.
- Left and Right side Edges;
  - y component of Stokes velocity is free, x component is kept as zero;
  - Darcy velocity, which is understood as flux is also kept zero;
  - normal flux dof fixed with a zero flux.
- Top Edge;
  - Stokes velocity $\check{\mathbf{v}}_s = \mathbf{0}$;
  - Darcy velocity $\check{\mathbf{v}}_l = \mathbf{0}$;
  - Normal flux is also determined to be zero.

We assume no velocity in x direction, $\check{\mathbf{v}}_s \cdot \mathbf{i} = 0$, and $\check{\mathbf{v}}_r \cdot \mathbf{i} = 0$, where $\mathbf{i}$ is the unit vector in x direction pointing from left to right. Let $u = \phi \check{\mathbf{v}}_r \cdot \mathbf{j}$ and $v = \check{\mathbf{v}}_s \cdot \mathbf{j}$, where $\mathbf{j}$ is the unit vector in y direction pointing upwards.

$$\begin{aligned} u + \phi^2 q_l' &= 0, \\ u' + \phi(q_l - q_s) &= 0, \\ [q_s - \frac{1}{3}(1 - 4\phi)v']' &= 0, \\ v' - \phi(q_f - q_s) &= 0, \end{aligned} \tag{1.142}$$

When no melting presents $\phi = 0$, the equations reduce to

$$u = -v = 0, \quad \text{and} \quad q_s' = 1, \quad \text{and} \quad q_l = 0. \tag{1.143}$$

When melting presents $\phi > 0$, the equations can be reduced to

$$\phi^{2+2\Theta}\left(\frac{3+\phi-4\phi^2}{3\phi}u'\right)' - u = \phi^{2+2\Theta}(1-\phi), \tag{1.144}$$

where all other quantities can be determined according to u as

$$v = -u, \quad \text{and} \quad q_l' = -\phi^{-2-2\Theta}u, \quad \text{and} \quad qs = u'/\phi + q_l. \tag{1.145}$$

When ϕ is constant, we define an auxiliary function R as

$$R = \left(\phi^{2+2\Theta}\frac{3+\phi-4\phi^2}{3\phi}\right)^{-1/2}. \tag{1.146}$$

We test our MFEM code on three porosity settings, and use the closed form solutions derived in Arbogast et al. (2017).

- Constant porosity;

$$\phi_0(z) = \phi_0; \tag{1.147}$$

  Closed form solution:

$$\begin{aligned}
u &= -\phi_0^2(1-\phi_0)\left[1 - \frac{\cosh(Rz)}{\cosh(RL)}\right],\\
v &= -u,\\
q_l &= (1-\phi_0)\left[z - \frac{1}{R}\frac{\sinh(Rz)}{\cosh(RL)}\right] + c,\\
q_s &= (1-\phi_0)\left[z - \frac{1-4\phi_0}{3+\phi_0-4\phi_0^2}\frac{\phi_0}{R}\frac{\sinh(Rz)}{\cosh(RL)}\right] + c,
\end{aligned} \tag{1.148}$$

  where c is a constant kernel.

- Piecewise constant porosity;

$$\phi_J(z) = \begin{cases} 0 & \text{if } z > 0, \\ \phi_0 & \text{if } z \le 0, \end{cases} \tag{1.149}$$

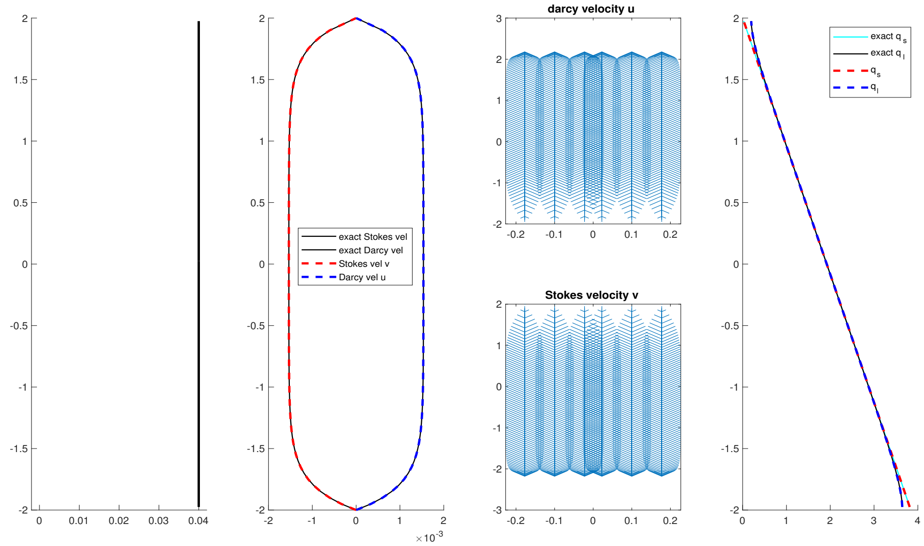


Figure 1.11: Constant porosity compaction column test.

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	3.100e-05	-	3.100e-05	-	3.285e-03	-	3.638e-05	-
2×40	8.047e-06	1.95	8.047e-06	1.95	8.731e-04	1.91	9.671e-06	1.91
2×80	2.031e-06	1.99	2.031e-06	1.99	2.218e-04	1.98	2.457e-06	1.98
2×160	5.091e-07	2.00	5.091e-07	2.00	5.567e-05	1.99	6.166e-07	1.99

Table 1.3: Constant porosity compaction column test.

Closed form solution:

We conclude that $u = -v = 0$ when $z > 0$, and $q_l = 0$, $q_s = z + c$. When $z < 0$, $\phi_0 \neq 0$,

$$\begin{aligned} u &= \phi_0^2(1-\phi_0)\left[1 - \cosh(Rz) + \frac{1-\cosh(RL)}{\sinh(RL)}\sinh(Rz)\right], \\ v &= -u, \\ q_l &= -(1-\phi_0)\left[z - \frac{1}{R}\sinh(Rz) + \frac{1-\cosh(RL)}{R\sinh(RL)}\cosh(Rz)\right] + c, \\ q_s &= q_l + \frac{u'}{\phi_0} + c, \end{aligned} \tag{1.150}$$

where c is a constant kernal shifting the pressure potential.

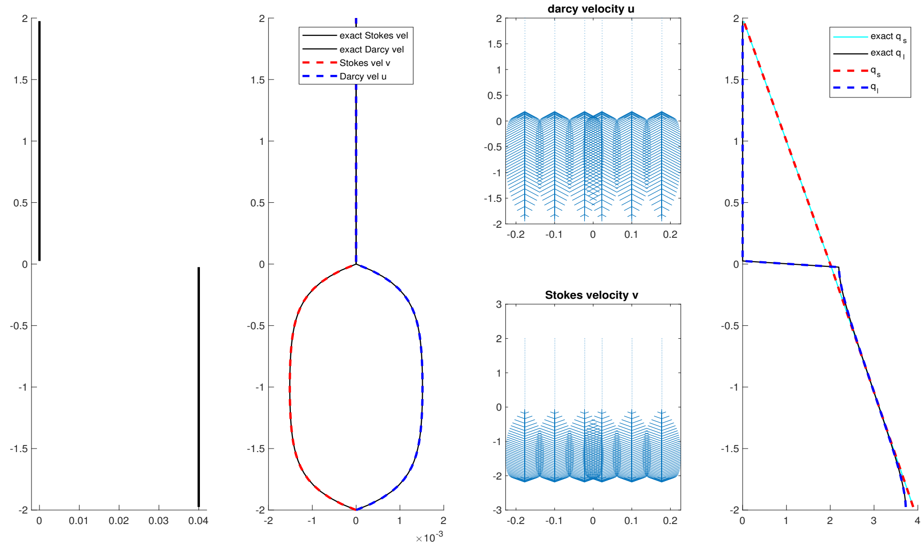


Figure 1.12: Piecewise constant porosity compaction column test.

	u		v		q_l		q_s	
$m \times n$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	2.980e-05	-	2.980e-05	-	3.794e-03	-	4.202e-05	-
2×40	7.681e-06	1.96	7.681e-06	1.96	1.005e-03	1.92	1.114e-05	1.92
2×80	1.958e-06	1.97	1.958e-06	1.97	2.472e-04	2.02	2.738e-06	2.02
2×160	4.974e-07	1.98	4.974e-07	1.98	5.972e-05	2.04	6.615e-07	2.05

Table 1.4: Constant porosity compaction column test.

- Quadratic porosity.

$$\phi_2(z) = \begin{cases} 0 & \text{if } \ z > 0, \\ \phi_0 z^2 & \text{if } \ z \leq 0, \end{cases} \tag{1.151}$$

Closed form solution:

We conclude that $u = -v = 0$ when $z > 0$.

$$\begin{aligned} u &= \frac{\phi_0^2}{1 - 4\phi_0}\left(L^{4-r_1} z^{r_1} - z^4\right), \\ v &= -v \\ q_l &= \frac{1}{1 - 4\phi_0}\left(z + \frac{L^{4-r_1}(-z)^{r_1-3}}{r_1 - 3}\right) + c, \\ q_s &= z + c, \end{aligned} \tag{1.152}$$

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	9.616e-07	-	9.616e-07	-				
2×40	3.212e-07	1.58	3.212e-07	1.58				
2×80	9.028e-08	1.83	9.028e-08	1.83				
2×160	2.746e-08	1.72	2.746e-08	1.72				

Table 1.5: Constant porosity compaction column test.

where

$$r_1 = \frac{3 + \sqrt{9 + 4/\phi_0}}{2}. \tag{1.153}$$

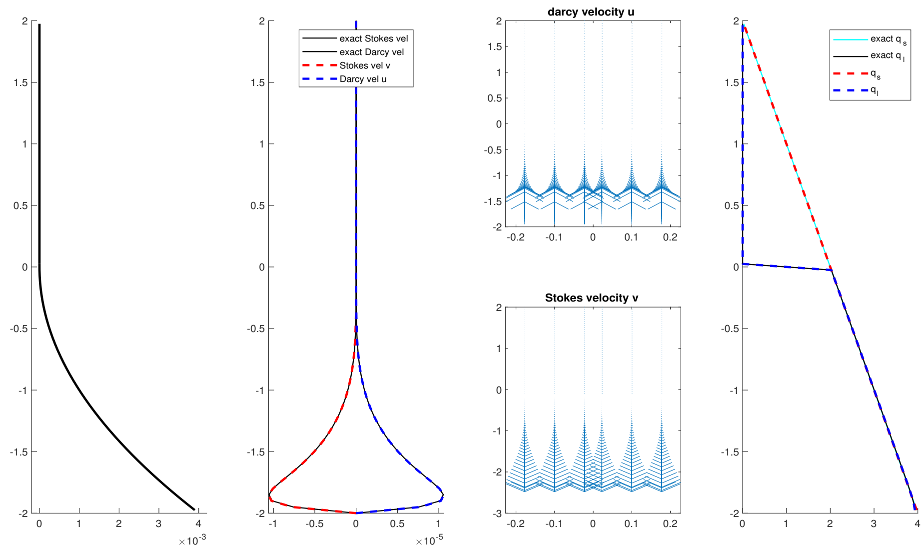


Figure 1.13: Quadratic porosity compaction column test.

Problem Setup 3: Mid-Ocean Ridge.

We now redo the Mid-Ocean ridge test case in Arbogast et al. (2017). We solve the full system of equations on the rectangular domain $-160\text{km} < x < 160\text{km}$, $0, x, 160\text{km}$. The boundary conditions are imposed by the constant porosity corner flow solution of Stokes velocity as

$$\mathbf{v}_s = \frac{2U_0}{\pi(x^2 + z^2)} \begin{pmatrix} \tan^{-1}(x/z)(x^2 + z^2) - xz \\ -z^2 \end{pmatrix}, \tag{1.154}$$

where $U_0 = 10^{-9}\text{m/s} = 3.2\text{cm/yr}$ is the constant upwelling velocity of mantle. According to mass conservation condition, we set no Darcy flux across the boundary. We then set a no identically vanishing porosity by the formula

$$\phi(x,z) = \begin{cases} 0.05(\dfrac{120\text{km} - z}{120\text{km}})^2(1 - \dfrac{|x|}{z+l}), & \text{if } x \leq 120\text{km and } |x| \leq z + l, \\ 0, & \text{otherwise,} \end{cases} \tag{1.155}$$

where $l = 20\text{m}$ is the offset set to avoid singularity computation. at the center $x = 0$ we translate the coordinates x to $x - l$ if $x < 0$, and $x + l$ when $x > 0$.

1.7 Finite Volume Method

The finite volume method (FVM) is a discretization method widely used in various scenarios that conservation laws are involved. The finite volume is a robust algorithm that can be applied to arbitriary complex geometry. The most important aspect of FVM that benefits this simulation is that the local mass conservation is achieved naturally. To begin with, we consider a general conservation law retarding a scalar variable u as

$$\begin{cases} \dfrac{\partial u(\mathbf{x},t)}{\partial t} + \nabla \cdot \mathbf{f}(u;\mathbf{x},t) = 0, & t > 0, \mathbf{x} \in \Omega, \\ u(\mathbf{x},0) = u_0(\mathbf{x}), & \mathbf{x} \in \Omega, \end{cases} \tag{1.156}$$

where we assume the domain $\Omega \subset \mathbb{R}^2$. The term $\mathbf{f}$ is a flux transporting scalar variable u. We write the basic semi-discretized form of basic finite volume method as

$$\frac{\partial \overline{u}}{\partial t} + \frac{1}{|E|}\int_{\partial E} \mathbf{F}(\mathbf{x},t) \cdot \boldsymbol{\nu} d\mathbf{x} = 0, \tag{1.157}$$

where we use the cell-averaged value $\overline{u} = \frac{1}{|E|}\int_E u(\mathbf{x},t)d\mathbf{x}$, and $F(\mathbf{x},t)$ is the approximated numerical flux. We use the Lax-Friedrichs numerical flux

$$\mathbf{F}(u^+,u^-) = \frac{1}{2}[(\mathbf{f}(u^+) + \mathbf{f}(u^-)) \cdot \boldsymbol{\nu} - \alpha_{LF}(u^+ - u^-)], \tag{1.158}$$

where u^+ and u^- represent values approximated from element $E^+ \in \mathcal{T}_h$ and $E^- \in \mathcal{T}_h$ respectively, and $\boldsymbol{\nu}$ is the unit vector pointing from E^+ to E^- orthogonal to the edge. Here, α_{LF} is the Lax-Fredrich stabilizer factor. We have two different ways to compute α_{LF} depending on the knowledge of solution.

- The global LF Stabilizer: $\alpha_{LF} = \max_u |\partial(\mathbf{f} \cdot \boldsymbol{\nu})/\partial u|$;
- The local LF Stabilizer: $\alpha_{LF} = \max(|\partial(\mathbf{f}^+ \cdot \boldsymbol{\nu})/\partial u^+|, |\partial(\mathbf{f}^- \cdot \boldsymbol{\nu})/\partial u^-|)$.

The numerical flux $\mathbf{F}$ (1.158) is consistent and conservtive across the edge. We now replace the u^+ and u^- with reconstructed values and rewrite the numerical flux as

$$\mathbf{F}(\mathcal{R}^+, \mathcal{R}-) = \frac{1}{2}[(\mathbf{f}(\mathcal{R}^+) + \mathbf{f}(\mathcal{R}^-) \cdot \boldsymbol{\nu} - \alpha_{LF}(\mathcal{R}^+ - \mathcal{R}^-)], \tag{1.159}$$

where $\mathcal{R}$ is a higher order reconstruction value. We evaluate the integrals over each edge $e_i \in \partial E$ with a quadrature rule. We represent length of edge e_i and area of cell E with $|e_i|$ and $|E|$ respectively. Assuming a quadrature rule with G points on each edge, we have

$$\overline{u}_t + \sum_{i=0}^{4} \frac{|e_i|}{|E|} \sum_g \omega_{i,g} \mathbf{F}(\mathcal{R}^+_{i,g}, \mathcal{R}^-_{i,g}) = 0, \tag{1.160}$$

where $\omega_{i,g}$ represents gth quadrature points on edge e_i.

1.8 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop a accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for reconstruction of discontinuous values with cell averaged values in Arbogast et al. (2024). The ML-WENO reconstruction will be primarily used in reconstruction of values on both sided of the edge for numerical flux computation appears in FVM discretization. It can also be applied to general reconstruction of discontinuous variables, e.g., discontinuous darcy pressure appears on the zero-nonzero porosity interface.

1.8.1 Stencil Polynomial

Given a stencil $\mathcal{S} = \{E_0, E_2, \dots, E_k\}$ of the quasiuniform logically rectangular mesh. We can define an associated tensor product stencil polynomial $Q_{s,t} \in \mathbb{Q}_{(s,t)} := \mathbb{P}_{s-1}(x) \otimes \mathbb{P}_{t-1}(y)$, where s and t equal to the stencil sizes in the x- and y-directions, respectively. We scale our stencil polynomial individually as

$$Q_{s,t} = \sum_{p<s,q<t} c_{p,q} \Big(\frac{x - x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p \Big(\frac{y - y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{1.161}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is the center point of the stencil $\Omega_{\mathcal{S}} = \cup_{E\in\mathcal{S}} \subset \mathbb{R}^2$, and $h_{\mathcal{S}} = \sqrt{\sum_k E_k/k}$ is the diameter of average cell size. Let $\xi = (x - x_{\mathcal{S}})/h_{\mathcal{S}}$ and $\eta = (y - y_{\mathcal{S}})/h_{\mathcal{S}}$ be the normalized varaibles, we can rewrite it into normalized version:

$$\hat{Q}_{s,t} = \sum_{p<s,q<t} c_{p,q} \xi^p \eta^q. \tag{1.162}$$

In finite volume scenerio, we are dealing with cell averaged value of the scalar transport variable

$$\overline{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}) d\mathbf{x} = \frac{1}{|E|} \int_E u(x, y) dx dy. \tag{1.163}$$

We can calculate coefficents $c_{p,q}$ by imposing a constraint that cell averaged value (1.163) is preserved through integral with stencil polynomial

$$\frac{1}{|E_k|} \int_{E_k} Q(\mathbf{x}) d\mathbf{x} = \overline{u}_{E_k}, \quad \forall E_k \in \mathcal{S}. \tag{1.164}$$

We will replace $\overline{u}_{E_k}$ with $\overline{u}_k$ in the following discussion for simplicity.

We can represent stencil polynomial with basis polynomials $Q_{s,t} = \sum_k \overline{u}_k \phi^k_{s,t}$, where basis polynomials $\phi^k_{s,t}$ can be calculated as

$$\frac{1}{|E_{k'}|} \int_{E_{k'}} \phi^k_{s,t}(\mathbf{x}) d\mathbf{x} = \frac{1}{|\hat{E}_{k'}|} \iint_{\hat{E}_{k'}} \hat{\phi}^k_{s,t}(\xi, \eta) d\xi d\eta = \begin{cases} 1 & k' = k \\ 0 & k' \neq k \end{cases} \quad \forall E_{k'}, E_k \in \mathcal{S}, \tag{1.165}$$

which would lead to a $(s \times t) \times (s \times t)$ non-singular linear system. We can now represent stencil polynomial with normalized basis polynomials as:

$$\hat{Q}_{s,t}(\xi, \eta) = \sum_k \overline{u}_k \hat{\phi}^k_{s,t}(\xi, \eta). \tag{1.166}$$

We define a linear projection operator $\pi_{\mathcal{S}}$ and let $\pi_{\mathcal{S}} u_{\mathcal{S}} = Q$ be a linear projection onto polynomial space W_p such that $\mathbb{Q}_{s,t} \subset W_p$. We estimate interpolation error with the Bramble-Hilbert Lemma Dupont and Scott (1980a) and Bramble and Hilbert (1970) in $H_r(\Omega)$ space. For simplicity, we assume $r = s = t$.

Lemma 1.12. *Let Ω_S be the domain covered by stencil $\mathcal{S}$ in $\mathbb{R}^2$ and let $u \in H^r(\Omega_{\mathcal{S}})$. We have:*

$$\|u - \pi_S u\|^2 \leq h^{2(r-1)} |u|^2_{H^r(\Omega_S)}. \tag{1.167}$$

Proof.

$$\|u - \pi_S u\| = \tag{1.168}$$

□

In a general 2D situtation when $s \neq t$, the exececssive order will be "wasted", since the order or accuracy will be $\min(s, t)$. However, in some cases, where we aquire the prior knowledge of the solution being qusi-1D, e.g., the transport velocity is fixed in one direction. We can have uneven stencils that has $\max(s, t)$ order of accuracy but in the desired direction. In other cases when derivative is involved, we also need an stretched stencil to compensate for the order loss when approximates derivative.

1.8.2 Stencil polynomial smoothness indicators

We define a smoothness indicator σ as a measurement of smoothness of $u(\mathbf{x})$ on the stencil S. The classic method from Jiang and Shu (1996) is defined in analogue to a cell-averaged Sobolev seminorm as:

$$\sigma_P = \sum_{1 \leq |\alpha| \leq m} \int\int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x, y))^2 dx dy, \tag{1.169}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a multi-index, $|\alpha| = \alpha_1 + \alpha_2$, and $\mathcal{D}$ is the derivative operator $\mathcal{D}^\alpha = \partial^{|\alpha|}/\partial x^{\alpha_1}\partial y^{\alpha_2}$. The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$
\begin{aligned}
\sigma_P &= \sum_{1\le|\alpha|\le m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x,y))^2 dxdy \\
&= \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_0} \big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi,\eta)\big)^2 d\xi d\eta \\
&= \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'.j'} \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi,\eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi,\eta) d\xi d\eta \\
&= \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'j'} \sigma_{ij}^{i'j'},
\end{aligned}
\tag{1.170}
$$

where $\sigma_{ij}^{i'j'}$ can be precomputed after mesh is established after mesh is established.

Alternatively, we can expand the Jiang-Shu smoothness indicator according to polynomial coefficients, which paves the way for further simplification. Recall the tensor product polynomial version (1.162),

$$
\sigma_P = \sum \tag{1.171}
$$

We observe that the smoothness indicator exhibit following convergence property that there exists some value $C > 0$ such that when $h \to 0^+$,

$$
\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(1) & \text{Not Smooth,} \end{cases} \tag{1.172}
$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

1.8.3 Two alternative smoothness indicator

We proposed two alternative ways of defining smoothness indicator in Huang et al. (2023) and Arbogast et al. (2025).

- *Simple smoothness indicator*

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1}\sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{1.173}$$

where $\overline{u}_0$ is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the computational cost and to deal with distorted stencil on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

- *Polynomial smoothness indicator*

1.8.4 ML-WENO weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\{\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \ldots, L, L \geq 0\}$. The reconstruction of u on target cell $E_{i,j}$ is a convex combination of stencil polynomials (1.166)

$$\mathcal{R}(\mathbf{x}) = \sum_l \tilde{\omega}_l Q_l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{1.174}$$

where $\sum_l \tilde{\omega}_l = 1$ is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily chosing a set of linear weights $\{\omega_l\}$ and scale with stencil

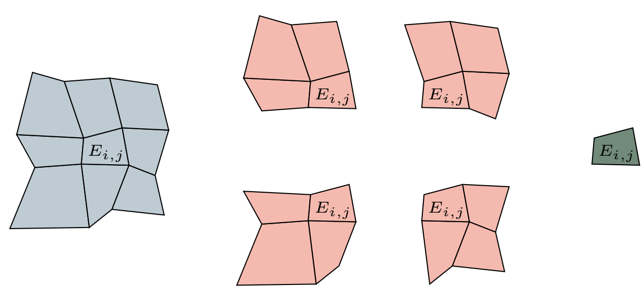


Figure 1.14: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{1.175}$$

where $r_l = \max\{s,t\}$, $\epsilon = 1e-4$ and $a > 0.5$ are picked as constants. The biasing factor $n_l = n(r_l)$ is used to biasing away further from constant level, whose smoothness indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{1.176}$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{1.177}$$

1.8.5 The ML-WENO Reconstruction Error

In this section, we consider the accuracy of our multi-level weighting scheme. We group the set of indices $l \in \mathcal{L} = \{0, 1, 2, \ldots L\}$ according to the order of reconstruction polynomials, i.e., the shape of individual stencil. We denote the set of indices of stencils have s cells in x direction and t cells in y direction as $\mathcal{L}_{s,t}$. We make the assumption that the mesh resolution is fine enough so that there exist at least one stencil that is smooth if shock presents. We define two sets holding indices of stencils with or without jump as

$$\begin{aligned} \text{Smooth} &= \{l : \quad u \text{ is smooth on } \mathcal{S}_l, \} \\ \text{Jump} &= \{l : \quad u \text{ has a jump on } \mathcal{S}_l\} \end{aligned} \tag{1.178}$$

where we assume Smooth $\cup$ Jump $= \mathcal{L}$ and Smooth $\cap$ Jump $= \emptyset$. We represent the set of indices of stencils of shape (s,t) that is smooth as $\mathcal{L}_{s,t} \cap$ Smooth, and the set of stencils of shat (s,t) that has a jump as $\mathcal{L}_{s,t} \cap$ Jump. For convience, we assume square stencils, where $r = s = t$. The accuracy result can be applied to $s \neq t$ situation directly in the selected direction. We show that the reconstruction is accurate with respect to the target cell E_0 to order

$$r_{\max} = \max\{r : \text{Smooth} \cap \mathcal{L}_{r,r} \neq \emptyset\}. \tag{1.179}$$

In Figure 1.15, we illustrate a (3,2,1) combination of stencils with thow shock presents. Here, Smooth $= \{0, 1\}$, which indicates that stencil $\mathcal{S}_0$ and $\mathcal{S}_1$ do not have jump present. In this case, we expect the accuracy of reconstruction is $r_{\max} = \text{Smooth} \cap \mathcal{L}_{2,2} = 2$.

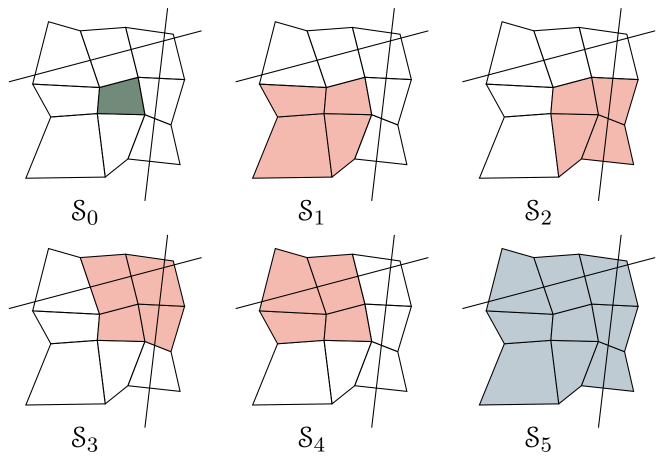


Figure 1.15: An illustration of a (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh with two shock presents.

We state the estimation of reconstruction error show in Arbogast et al. (2024). We start by estimating the error of our nonlinear scaling (1.174) asscoiated to some stencil $\mathcal{S}_l$ for some $m > 0$.

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m}. \tag{1.180}$$

We estimate the reconstruction error for two situtaions.

- $l \in$ Smooth, the smoothness indicator $\sigma_l = Ch^2 + \mathcal{O}(h^3)$;
- $l \in$ Jump, the smoothness indicator $\sigma_l = \Theta(1)$.

Use the asymptotic estimation result from Olver (1997). Let $n \geq m$, $a > 0$, and $b > 0$

$$\begin{aligned} \frac{a}{bh^m + \mathcal{O}(h^n)} &= \frac{ah^{-m}}{b}\big(1 + \mathcal{O}(h^{n-m})\big), \\ \frac{a}{bh^{-n} + \mathcal{O}(h^{-m})} &= \frac{ah^n}{b}\big(1 + \mathcal{O}(h^{n-m})\big). \end{aligned} \tag{1.181}$$

For the smooth stencil $\mathcal{S}_l \in \text{Smooth}$

$$
\begin{aligned}
\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m} &= \frac{\omega_l}{((C+\epsilon)h_0^2 + \mathcal{O}(h^3))^m} \\
&= \omega_l \Big(\frac{h_0^{-2}}{C+\epsilon}(1+\mathcal{O}(h)) \Big)^m \\
&= \frac{\omega_l}{(D+\epsilon)^m} h_0^{-2m}(1+\mathcal{O}(h)).
\end{aligned} \tag{1.182}
$$

In conclusion, we estimate the nonlinear scaling as

$$
\hat{\omega}_l = \begin{cases} \dfrac{\omega_l}{(C+\epsilon)^m} h_0^{-2m}(1+\mathcal{O}(h)), & \text{if} \quad l \in \text{Smooth}, \\ \Theta(1), & \text{if} \quad l \in \text{Jump}. \end{cases} \tag{1.183}
$$

Then we estimate the error for the normalized nonlinear weighting (1.176). We expand the sum of nonlinear scaling as

$$
\sum_l \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l + \sum_{l \in \text{Jump}} \hat{\omega}_l. \tag{1.184}
$$

We estimate the smooth part of the nonlinear weighting as

$$
\sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \frac{\omega_l}{(C+\epsilon)^m} h_0^{-2(ar+\eta)}(1+\mathcal{O}(h)). \tag{1.185}
$$

Assume $h < 1$, we keep the largest $ar + \eta(r)$ number as our estimator

$$
\begin{aligned}
\sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = & \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r_{\max}, r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))}(1+\mathcal{O}(h)) \\
& \qquad + \Theta(h^{-2(ar+\eta(r))}) \\
= & \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r_{\max}, r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))} \\
& \qquad + \mathcal{O}(h^{1-2(ar_{\max}+\eta(r_{\max}))}).
\end{aligned} \tag{1.186}
$$

later, we denote the set $\text{Smooth} \cap \mathcal{L}_r$ as Smooth_r for simplicity. The overall accuracy estimation of the nonlinear weighting (1.176) is presented as

- $l \in$ Smooth

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}} = \frac{\omega_l}{\sum_{l' \in \text{Smooth}_r} \omega_{l'}} h^{2[a(r_{\max}-r)+\eta(r_{\max})-\eta(r)]}(1+\mathcal{O}(h)); \tag{1.187}$$

- $l \in$ Jump

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}} = \Theta(h^{2[a(r_{\max})+\eta(r_{\max})]}). \tag{1.188}$$

Now we can estimate the reconstruction error as

$$\begin{aligned}
|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| = |\sum_l \tilde{\omega}_l (u(\mathbf{x} - Q_l(\mathbf{x})))| \\
&\leq \sum_{l \in \text{Smooth}} \tilde{\omega}_l |u(\mathbf{x}) - Q_l(\mathbf{x})| + \sum_{l \in \text{Jump}} \tilde{\omega}_l |u(\mathbf{x}) - Q_l(\mathbf{x})| \\
&\leq \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \Theta(h^{2[a(r_{\max}-r)+(\eta(r_{\max})-\eta(r))]})\mathcal{O}(h^r) \\
&\quad + \sum_{l \in \text{Jump}} \Theta(h^{2(ar_{\max}+\eta(r_{\max}))})\mathcal{O}(1) \\
&= \mathcal{O}(h^{r_{\max}})
\end{aligned} \tag{1.189}$$

Combining everything above and (1.173), we arrive at the formal error estimate of the reconstruction.

Lemma 1.13. *Let $\mathcal{R}(\mathbf{x})$ be the reconstruction. Assuming a qusi-uniform mesh, $h_0 = \Theta(h)$, there exists a constant $C > 0$ such that for any $\mathbf{x} \in E_0$,*

$$|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| \leq Ch^{r_{\max}}, \tag{1.190}$$

where $r_{\max}$ is defined by (1.178).

1.8.6 Handling Boundary Conditions

In our simulation, we couple FEM and FVM solver on a single mesh. Therefore, accurate and compatible boundary condition should be assigned accordingly. However, it is usually difficult to manage boundary condition for higher order finite volume and finite difference method, which require wide stencils. Some deal with the difficulty use ghost cells or points outside the boundary, and maintain the same stencil as if they were in the interior of the domain. However, the extrapolation or estimation of the manufactured values of ghost cells or ghost points are not straightforward. A notable method, the inverse Lax-Wendroff procedure, substitute the normal derivative of the solution with tangential and time derivatives using the PDE relation in an iteratvie way to impose more precise values for the ghost cells or points. The key to the above strategy is to have accurate and stable reconstruction near the boundary. ML-WENO gives the flexibility to add stenncils of the desired size that span back into the domain.

In order to assign proper boundary condition to a FVM discretization, we should determine whether the flow is inflow or outflow first. If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} > 0$, we recognize that it is a outflow boundary, therefore, no fixed boundary condition should be assigned. We use "free outflow" boundary that the numerical flux is simply $\mathbf{f}(u_B) \cdot \boldsymbol{\nu}$.

The inflow boundary can be understood in various ways. We can understand it as prescribing the inflow flux $\mathbf{f}_B$ or fixing the boundary value u_B as an equivelent to Dirichlet boundary in the context of Finite Element. In the special case when a zero flux is prescribed, we would call it noflow boundary. In the inflow case, there may be a discontinuous inflow shock wave. We require a swift response inside the domain to align with the specified Dirichlet value. We therefore add an extra constant stencil polynomial to the inflow boundary to accommodate the potential discontinuity.

There are some other special cases that require special treatment on the boundary, e.g., when symmetry should be enforced.

1.8.7 Numerical Test

For completeness, we test our ML-WENO algorithm with simple Riemann shock and rarefraction. Let $\mathbf{f}(u) = (u^2/2, u^2/2)$ be the Burgers flux function. In this example, we take the initial condition $u_0(x, y) = 1$, and the computational domain

$\Omega = [0, 3] \times [0, 1]$. The true solution is

$$u(x, y, t) = \begin{cases} 0, & x < 0.5, x \geq 0.5t + 1.5, \\ (x - 0.5)/t, & 0.5 \leq x < t + 0.5, \\ 1, & t + 0.5 \leq x < 0.5t + 1.5. \end{cases} \tag{1.191}$$

We forward martching to the time $= 2.0$, when trailing rarefaction reaches the leading shock. We use SSP2RK time stepping scheme.

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